

滤波器和谐振电路

必修8

第14章

14.3 谐振的定义和条件

14.4 串联谐振

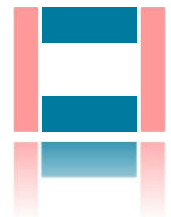
14.5 并联谐振

主讲人：邹建龙

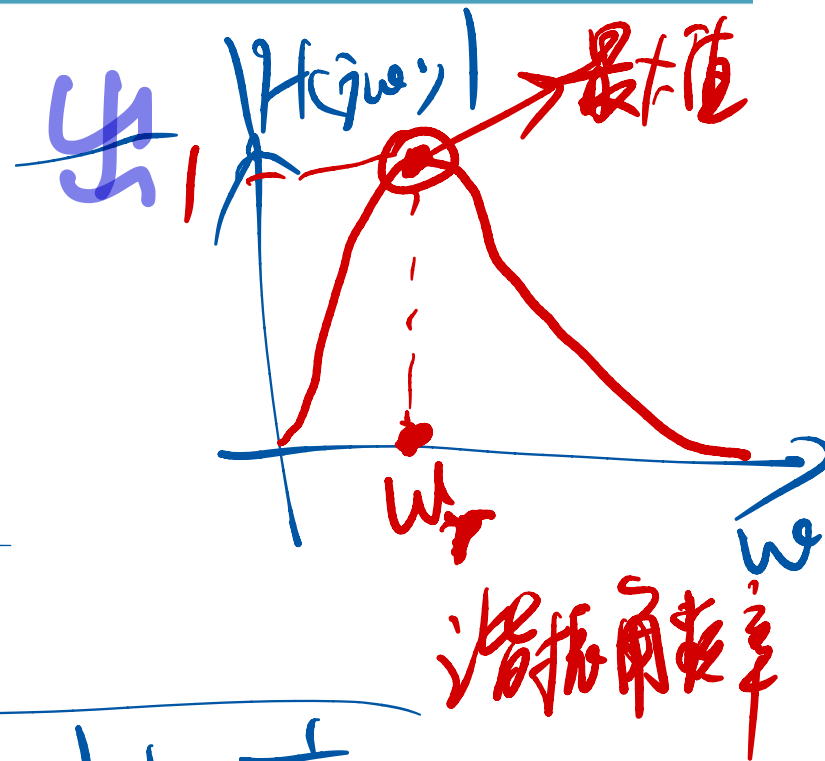
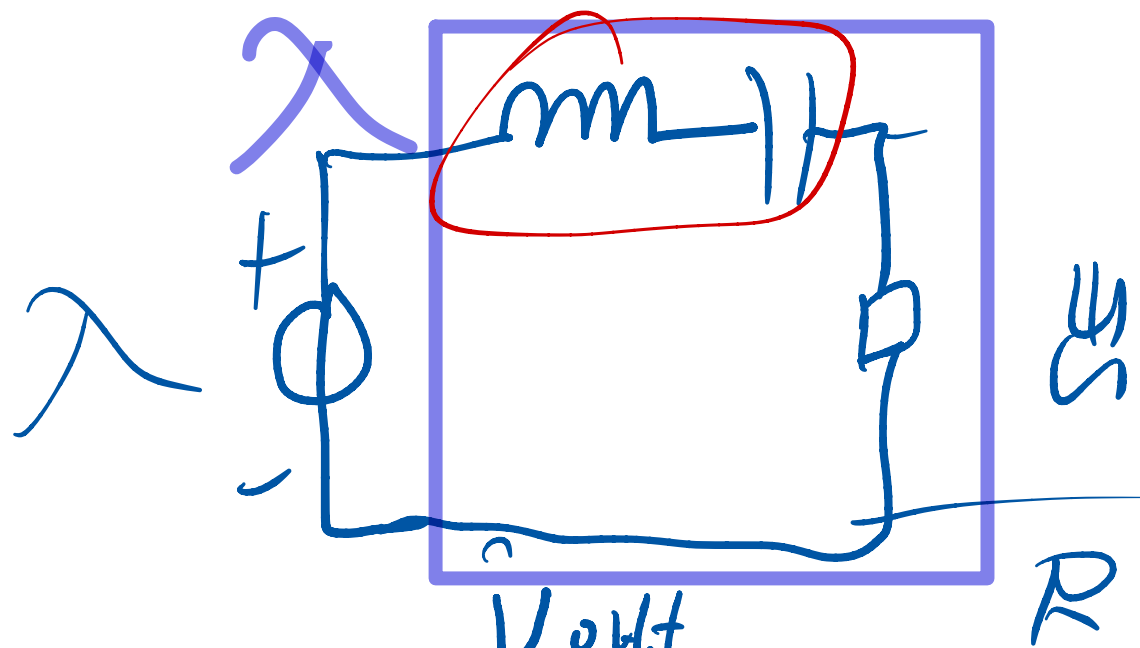
时 间：2025年5月26日



- 引言
- 14.3 谐振的定义和条件
- 14.4 串联谐振
- 14.5 并联谐振
- 小结

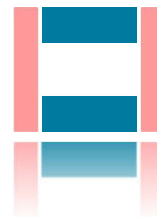


14.3-14.5 谐振——引言



$$H(j\omega) = \frac{U_{out}}{U_{in}} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}}$$

$$|H(j\omega)| = \frac{R}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$



14.3 谐振的定义和条件

谐振的定义：

② U 与 I 同相

③ 端口内 $X_{eq} = 0 \rightarrow$ 纯电阻

谐振 (resonance) 是含有电感和电容电路的一种特殊工作状态，在该工作状态下，电路的等效阻抗的虚部为零，即等效阻抗为纯电阻。

谐振的条件：

ω

满足谐振定义时，电路会发生谐振，也就是说，

谐振的定义就是谐振的条件。

阻抗虚部为0 $\text{Im}(Z_{eq}) = 0$

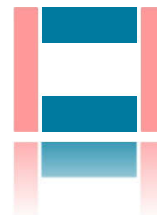
导纳 $\text{Im}(Y_{eq}) = 0$
 $\downarrow$ 在复数计算方便

$$Z_{eq} = R + j\omega L + \frac{1}{j\omega C}$$

$$= R + j(\omega L - \frac{1}{\omega C}) = R$$

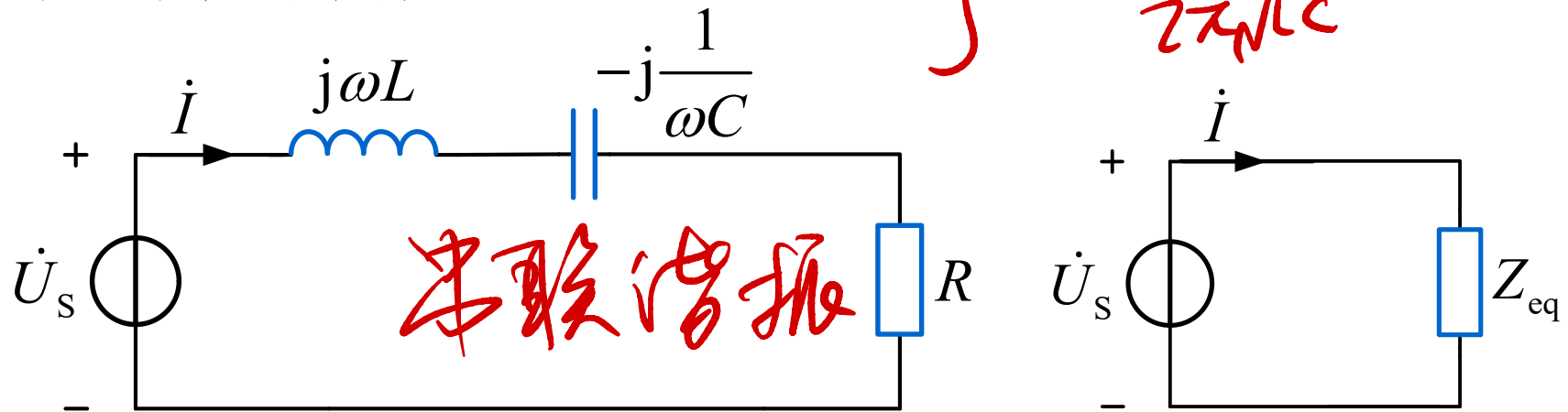
$$Y_{eq} = \frac{1}{Z_{eq}}$$

R



14.4 串联谐振

串联谐振的条件



$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$Z_{eq} = R + j\omega L - j\frac{1}{\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right) = 0$$

串联谐振条件: $\text{Im}(Z_{eq}) = 0$

$$\Rightarrow \omega L - \frac{1}{\omega C} = 0 \\ \omega L = \frac{1}{\omega C}$$

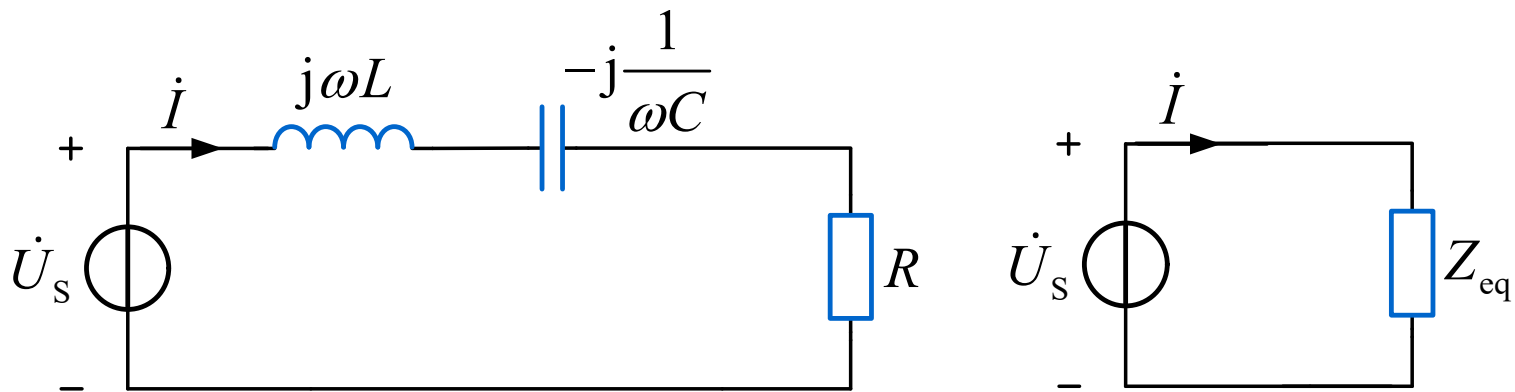
$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}, \text{ 称为串联谐振角频率!}$$

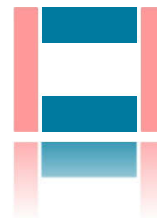
$\omega^2 LC = 1 \quad \omega = \frac{1}{\sqrt{LC}}$

resonance

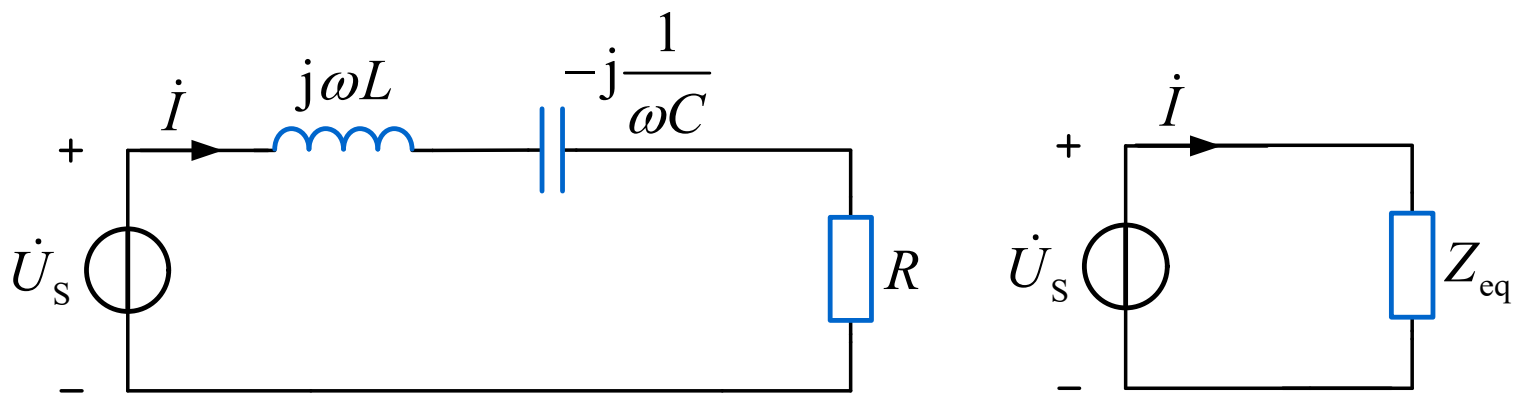
14.4 串联谐振



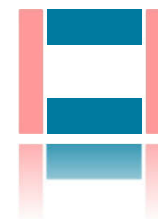
串联谐振的特点：



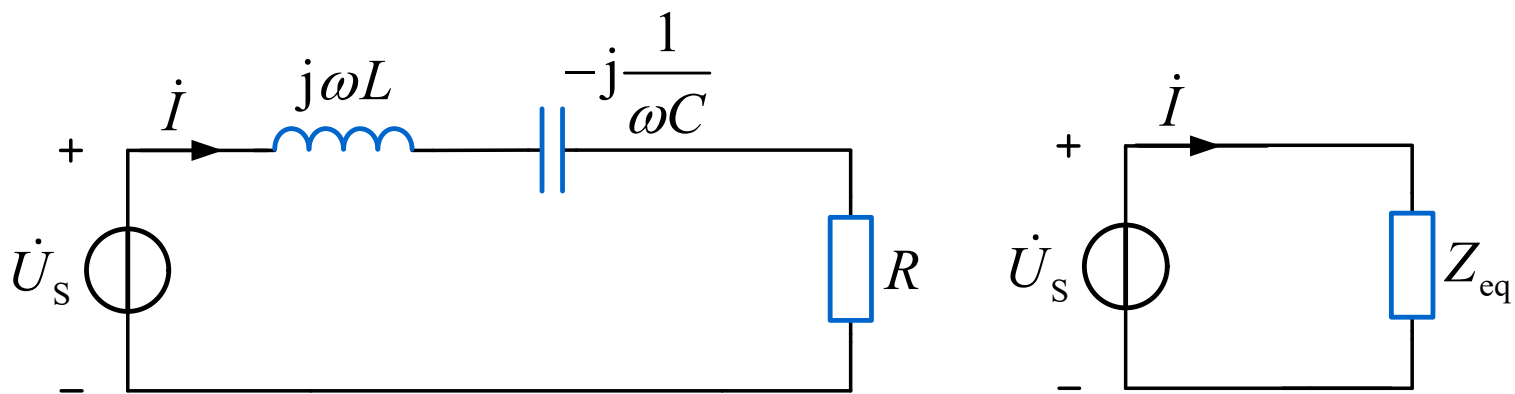
14.4 串联谐振



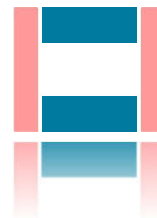
串联谐振的特点：



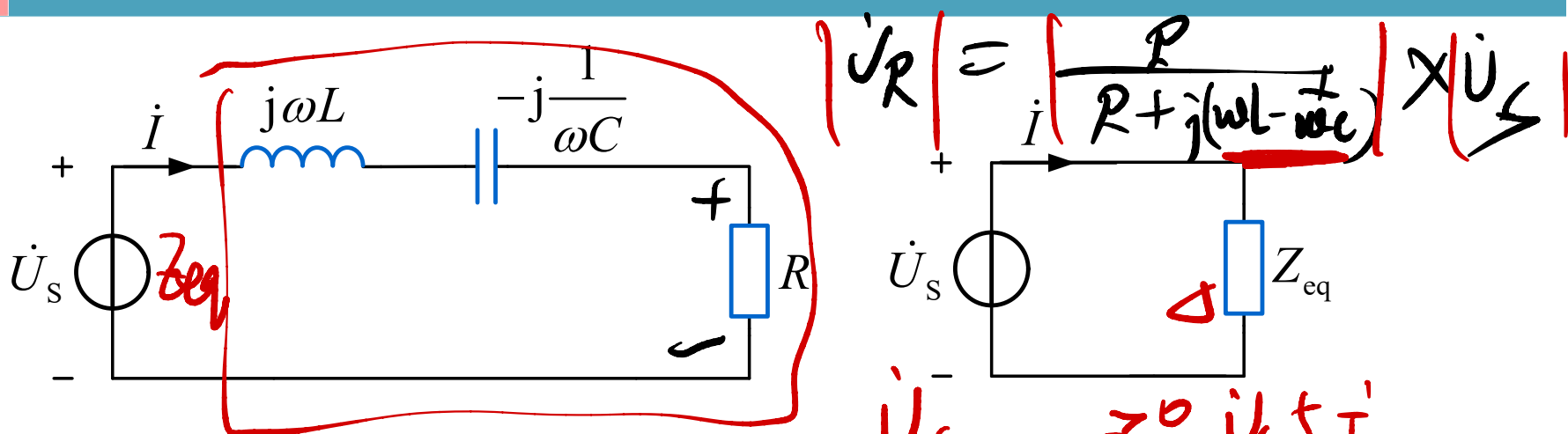
14.4 串联谐振



串联谐振的特点：



14.4 串联谐振



串联谐振的特点:

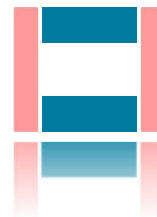
(1) 在关联参考方向下, 等效阻抗 (为纯电阻) 的 电压与电流同相位, 电路功率因数为1, 无功功率为0

(2) 等效阻抗的模值最小 $|Z|_{\min} = R$

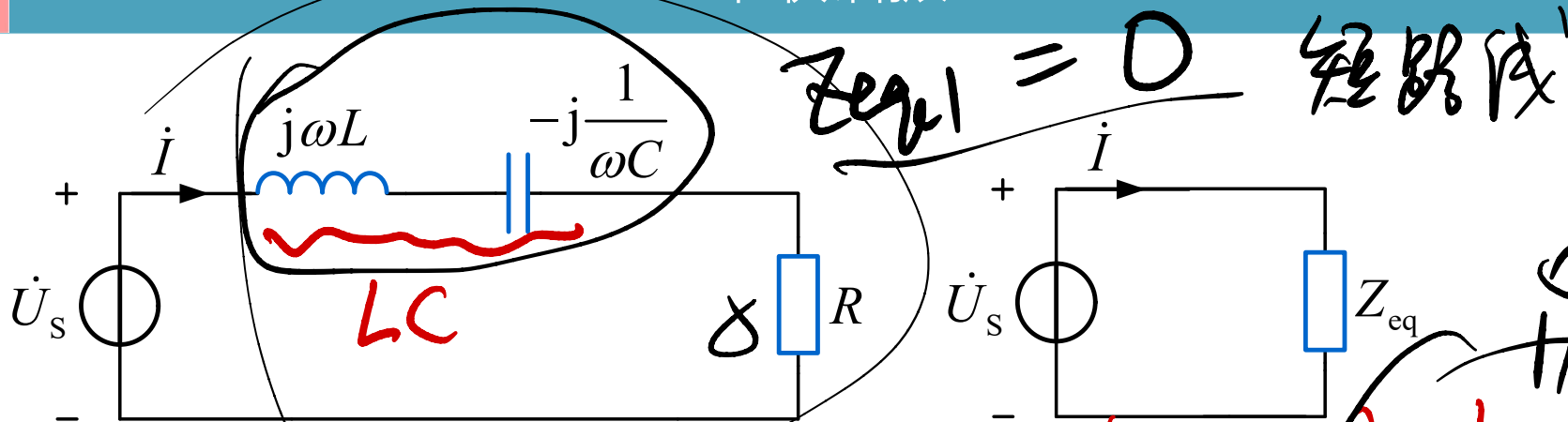
(3) 电阻电压有效值最大 $U_{R\max} = \frac{RU_s}{|Z|_{\min}} = \frac{RU_s}{R} = U_s =$

$$|Z_{eq}| = |R + j(\omega L - \frac{1}{\omega C})| = \sqrt{R^2 + (\Delta)^2} \geq 0$$

= 0 谐振



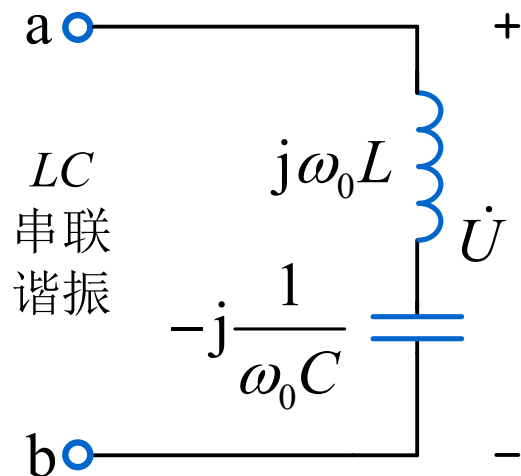
14.4 串联谐振



串联谐振的特点:

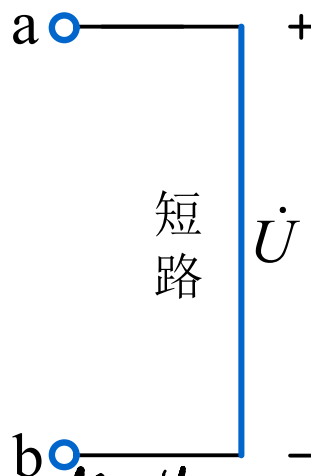
$$Z_{eq} = R + j\omega L + \frac{1}{j\omega C} = R + j(\omega L - \frac{1}{\omega C})$$

(4) LC串联谐振的等效阻抗为0, 相当于短路



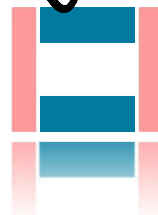
特殊

有用 $\rightarrow$ 滤波

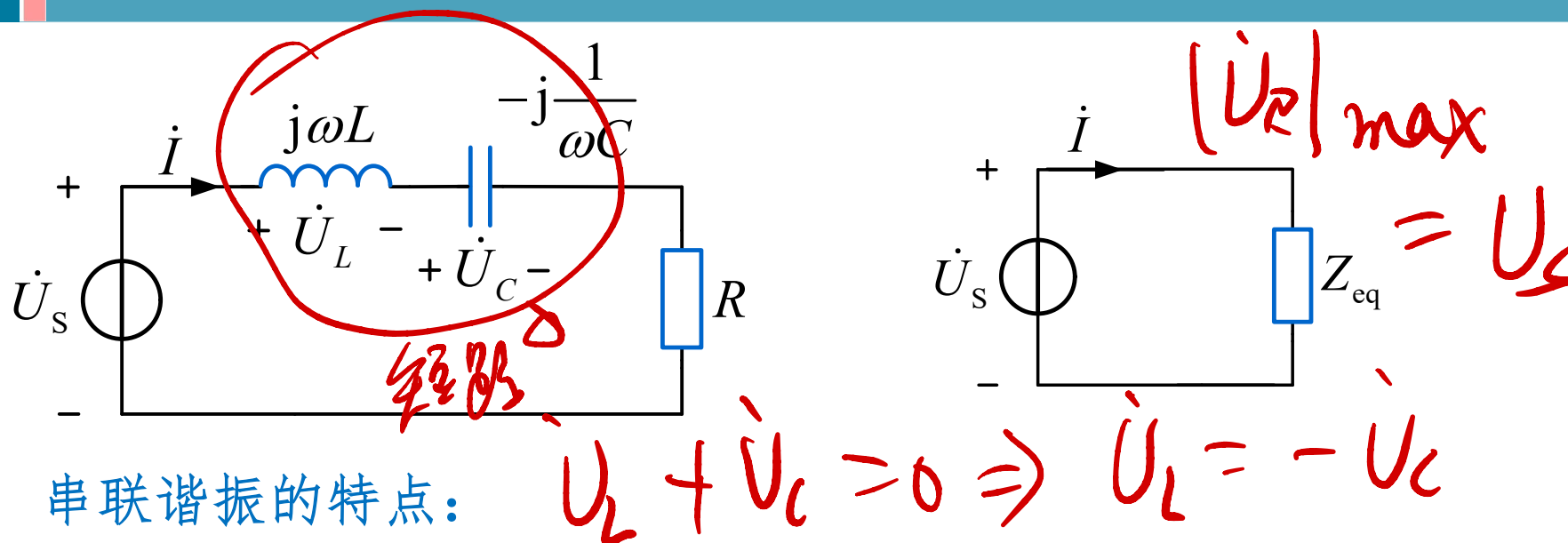


$$Z_{eq} = R$$

$$j\omega L + \frac{1}{j\omega C} = 0$$



14.4 串联谐振

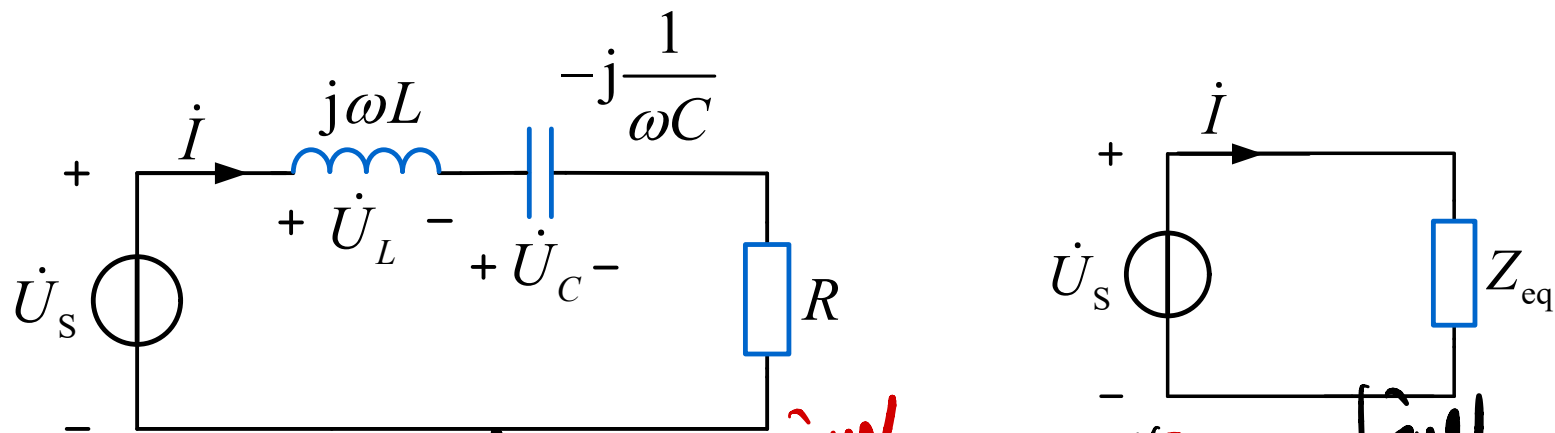


(5) 电感电压和电容电压互为相反数，有效值相等，可能超过电压源电压有效值。

$$\dot{U}_L + \dot{U}_C = 0 \quad U_L = U_C = \left| \frac{j\omega_0 L}{Z} \dot{U}_s \right| = \frac{\omega_0 L}{R} U_s = \left| \frac{1}{\frac{j\omega_0 C}{Z}} \dot{U}_s \right| = \frac{1}{R\omega_0 C} U_s$$

当品质因数 $Q = \frac{\omega_0 L}{R} = \frac{1}{R\omega_0 C} > 1$ 时，电容和电感出现过电压

14.4 串联谐振



串联谐振的特点: $\dot{U}_L = \frac{j\omega L}{R + j\omega L + \frac{1}{j\omega C}} \dot{U}_s$ $|\dot{U}_L| = \left| \frac{j\omega L}{R} \right| U_s$

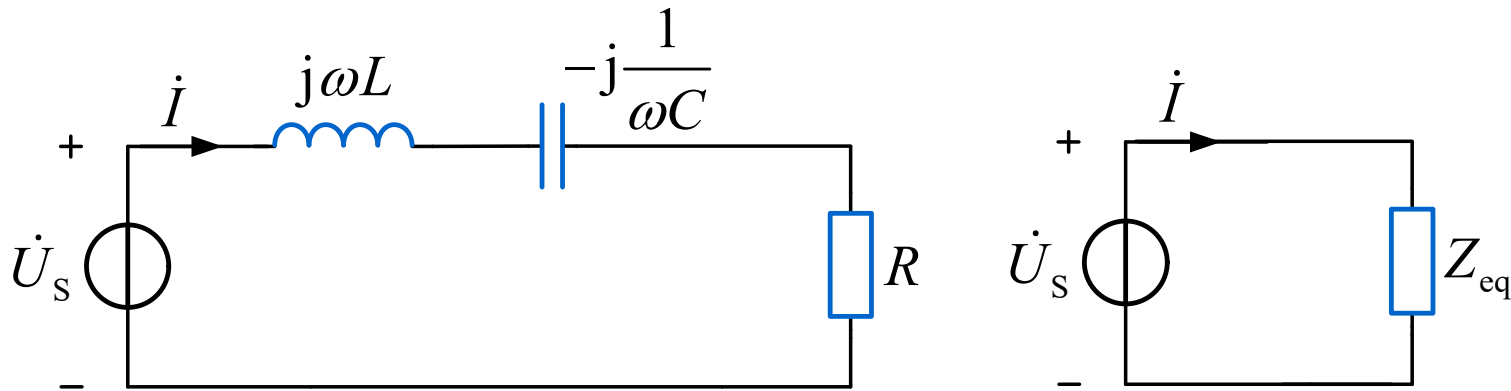
(5) 电感电压和电容电压互为相反数，有效值相等，可超过电压源电压有效值。

$$\dot{U}_L + \dot{U}_C = 0 \quad U_L = U_C = \left| \frac{j\omega_0 L}{Z} \dot{U}_s \right| = \frac{\omega_0 L}{R} U_s = \left| \frac{j\omega_0 C}{Z} \dot{U}_s \right| = \frac{U_L}{U_s} U_s = \frac{\omega L}{R} U_s$$

Quality Factor
 Δ

当品质因数 $Q = \left(\frac{\omega_0 L}{R} \right) = \left(\frac{1}{R\omega_0 C} \right) > 1$ 时，电容和电感出现过电压

14.4 串联谐振



串联谐振电路的品质因数定义：电路发生串联谐振时，电路储存的最大能量与一个周期的之比。

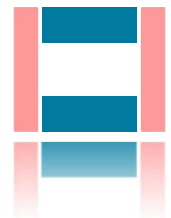
$$Q = 2\pi \frac{\text{最大储能}}{\text{一个周期耗能}} = \frac{\text{无功功率}}{\text{有功功率}}$$

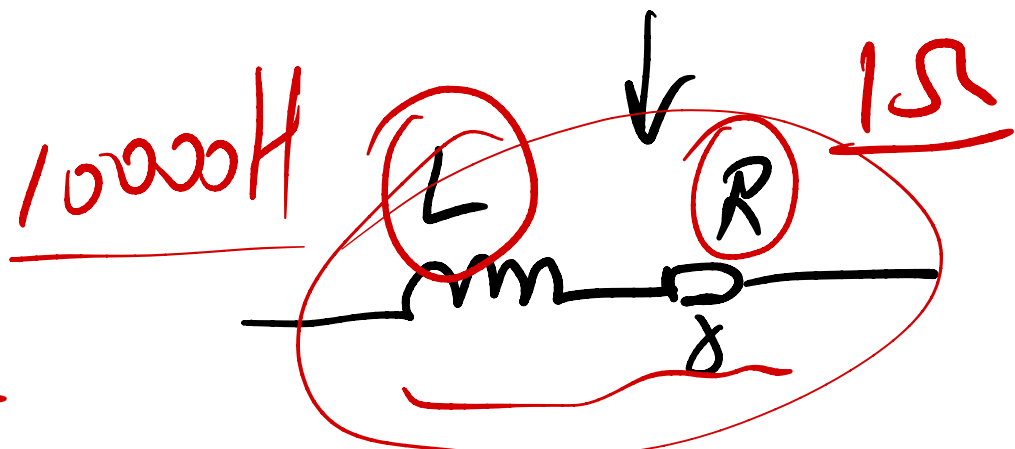
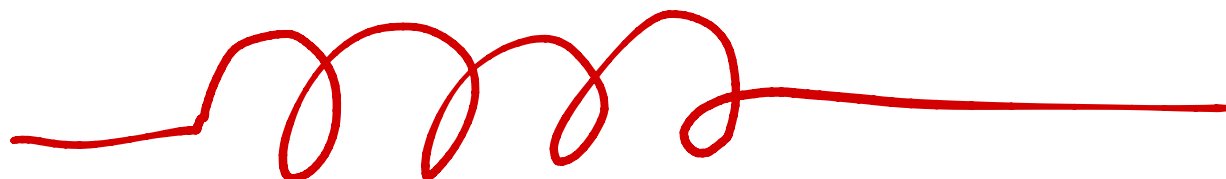
$$Q = 2\pi \frac{\frac{1}{2} L (\sqrt{2} I)^2}{R I^2 T} = \frac{2\pi f L}{R} = \frac{\omega_0 L}{R} = \frac{1}{R \omega_0 C}$$

电感 电容 电阻

$$= \frac{I^2 \omega L}{I^2 R}$$

电阻





$$R = \rho \frac{l}{S}$$

ω
R 影响

~~R 不是长~~

ω

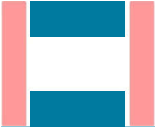
~~R 不是长~~

平衡

折中

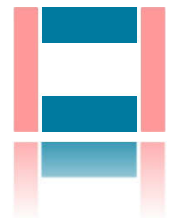
10⁻¹⁰⁰

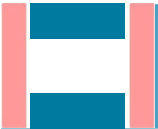
$$Q = \frac{\omega L_0}{\Delta R}$$



14.4 串联谐振

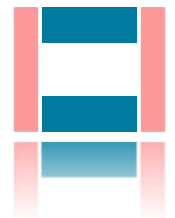
品质因数的由来





14.4 串联谐振

品质因数的由来

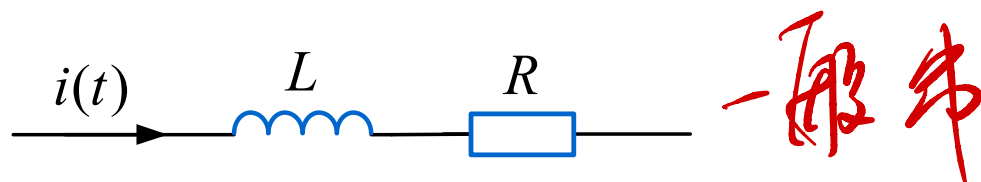


14.4 串联谐振

品质因数的由来

品质因数最初是用来定义实际电感器和实际电容器的品质

理想电感的作用是储能，而不是耗能，但实际电感器都有电阻，为了表征实际电感器的品质好坏，将储能与耗能之比定义为品质因数



$$Q = 2\pi \frac{\text{最大储能}}{\text{一个周期耗能}} = \left| \frac{\text{无功功率}}{\text{有功功率}} \right| = \frac{I^2 \omega L}{I^2 R} = \frac{\omega L}{R}$$

实际电感器也可以用电感和电阻并联建模，此时品质因数为

$$Q = \left| \frac{\text{无功功率}}{\text{有功功率}} \right| = \frac{U^2}{I^2 R} = \frac{1}{R \omega L}$$

14.4 串联谐振

品质因数的由来

品质因数最初是用来定义实际电感器和实际电容器的品质

理想电容的作用是储能，而不是耗能，但实际电容器都有电阻，为了表征实际电容器的品质好坏，将储能与耗能之比定义为品质因数

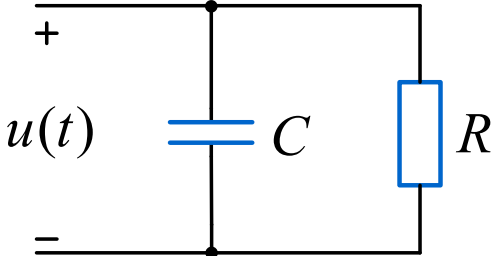


A circuit diagram showing a series connection of a capacitor C and a resistor R . An arrow labeled $i(t)$ indicates the current flowing through the circuit.

$$Q = 2\pi \frac{\text{最大储能}}{\text{一个周期耗能}} = \left| \frac{\text{无功功率}}{\text{有功功率}} \right| = \left| \frac{-I^2 \frac{1}{\omega C}}{I^2 R} \right| = \frac{1}{R\omega C}$$

串

实际电容器也可以用电容和电阻并联建模，此时品质因数为



A circuit diagram showing a parallel connection of a capacitor C and a resistor R . The voltage across the parallel combination is labeled $u(t)$ with a '+' sign at the top and a '-' sign at the bottom.

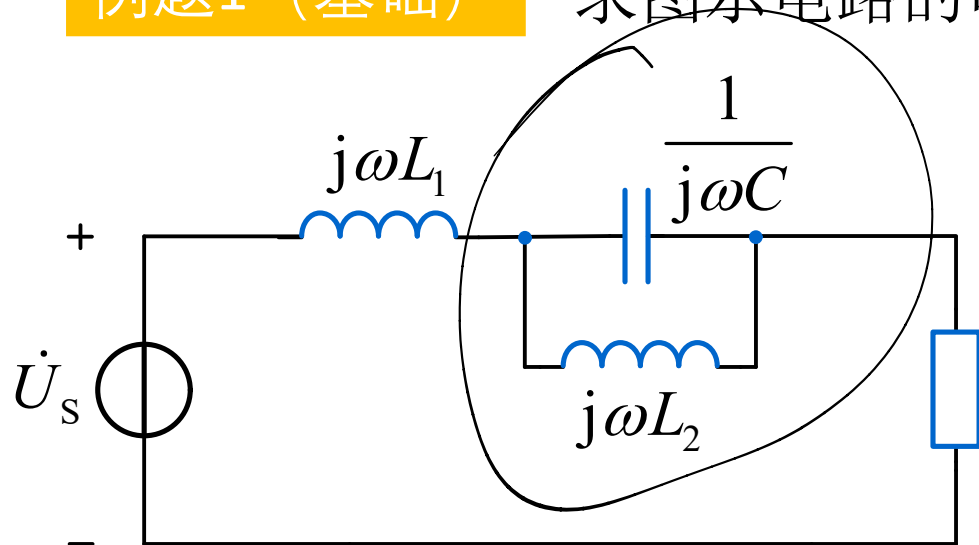
$$Q = \left| \frac{\text{无功功率}}{\text{有功功率}} \right| = \frac{\omega C U^2}{\frac{U^2}{R}} = R\omega C$$

并

14.4 串联谐振

例题1 (基础)

求图示电路的串联谐振角频率



串联谐振

$$Z_m(z_{eq}) = 0$$

$$j\omega L_1 + j\omega L_2$$

$$z_{eq} = R + j\omega L_1 + \frac{1}{\frac{1}{j\omega C} + j\omega L_2}$$

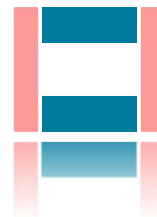
$$= R + j(\quad)$$

$$Z_m(z_{eq}) = X = 0$$

$$\omega \Rightarrow \omega$$

$$Z_m(z_{eq}) = 0 \rightarrow \text{串}$$

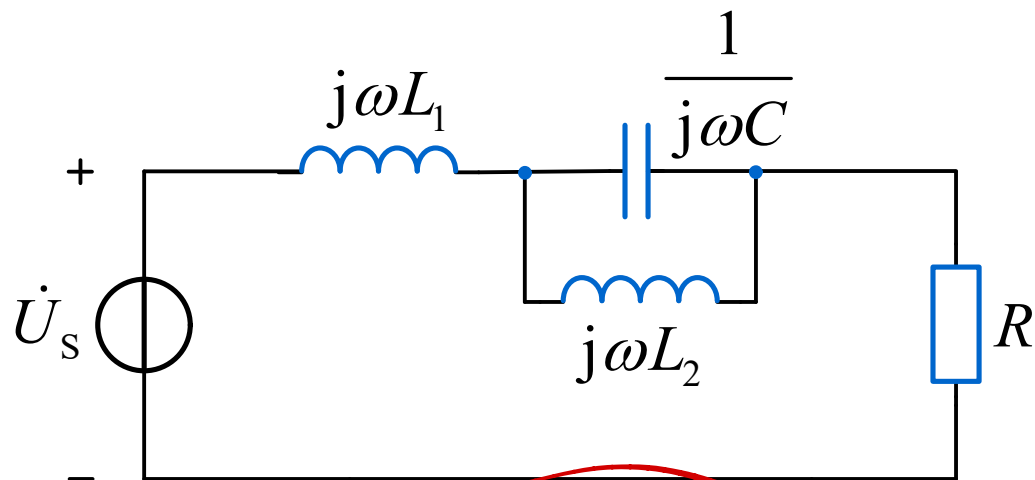
$$Z_m(y_{eq}) = 0 \rightarrow \text{并}$$



14.4 串联谐振

例题1 (基础)

求图示电路的串联谐振角频率



$$Z_{\text{eq}} = R + j\omega L_1 + \frac{j\omega L_2 \times \frac{1}{j\omega C}}{j\omega L_2 + \frac{1}{j\omega C}}$$

$\Rightarrow 0$

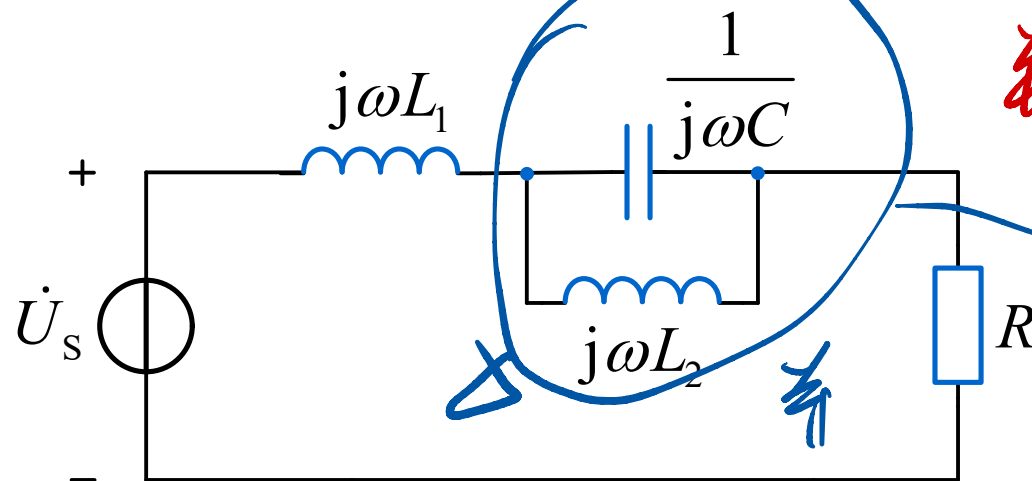
$$\text{Im}(Z_{\text{eq}}) = 0$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{\frac{L_1 L_2}{L_1 + L_2} C}}$$

14.4 串联谐振

例题1 (基础)

求图示电路的串联谐振角频率



$$Z_{eq} = R + j\omega L_1 + \frac{j\omega L_2 \times \frac{1}{j\omega C}}{j\omega L_2 + \frac{1}{j\omega C}}$$

$$\text{Im}(Z_{eq}) = 0$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{\frac{L_1 L_2}{L_1 + L_2} C}}$$

$$\text{Im}(Y_{eq}) = 0$$

$$\frac{1}{j\omega L_2} + j\omega C = 0$$

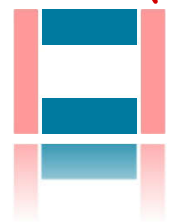
$$\omega = \sqrt{L_2 C}$$

$$Z_{eq} = 10$$

$$\omega = \sqrt{L_2 C}$$

$$\omega = \sqrt{L_2 C}$$

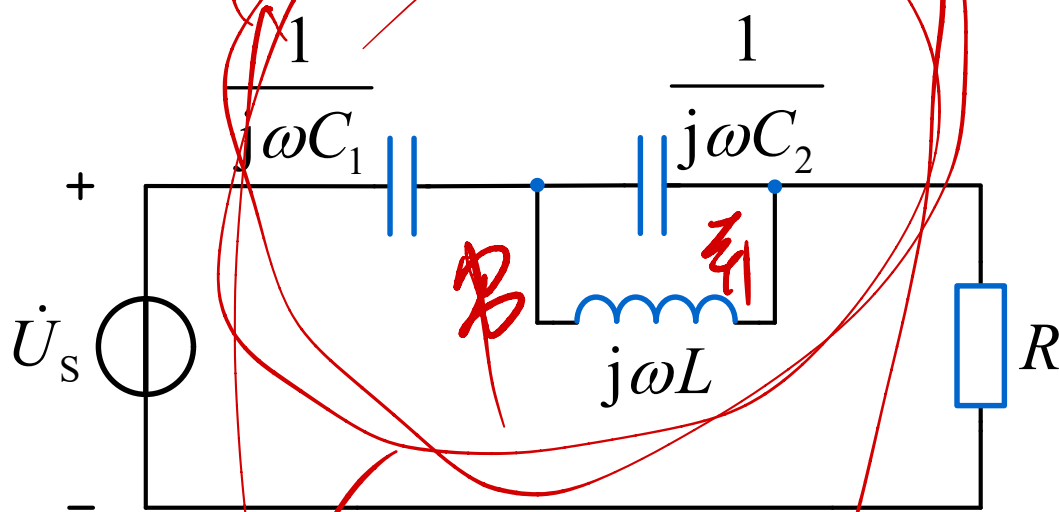
$$\omega = \sqrt{L_2 C}$$



14.4 串联谐振

同步练习题1 (基础)

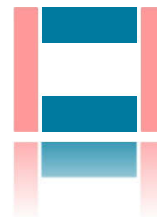
求图示电路的串联谐振角频率



$$Z_{eq1} = 0$$

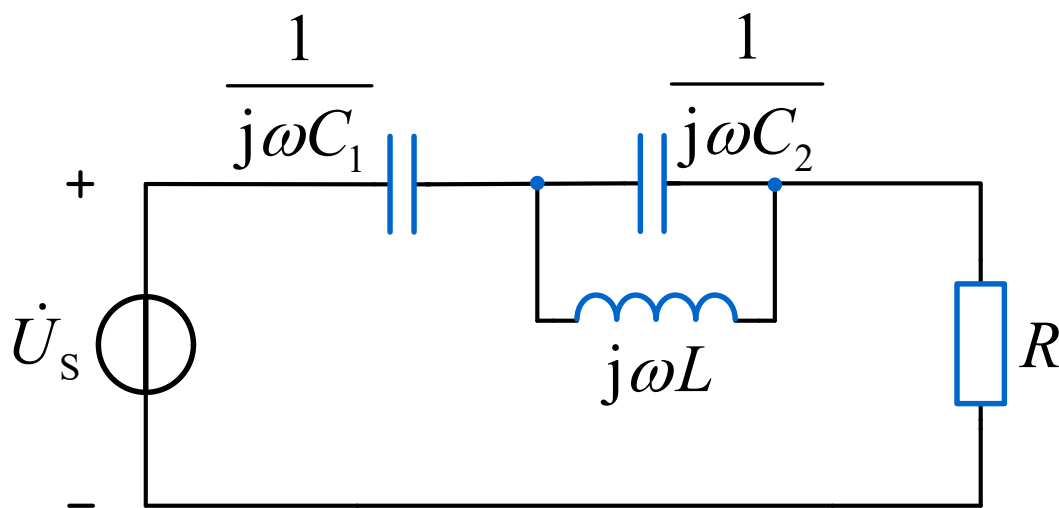
Z_{eq1}

$$\begin{aligned} \operatorname{Im}(Z_{eq}) &= \operatorname{Im}(R + jX) = 0 \\ &= X = 0 \end{aligned}$$



14.4 串联谐振

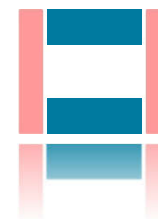
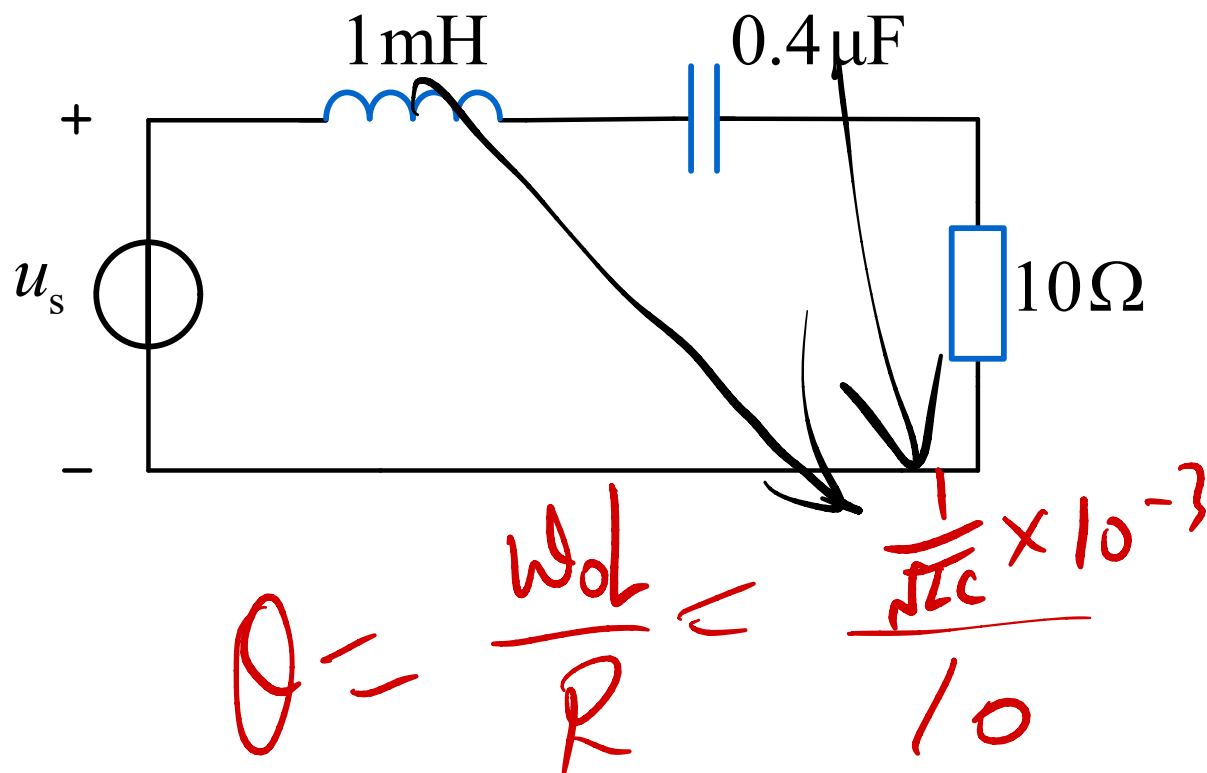
同步练习题1（基础） 求图示电路的串联谐振角频率



答案: $\omega_0 = \frac{1}{\sqrt{L(C_1 + C_2)}}$

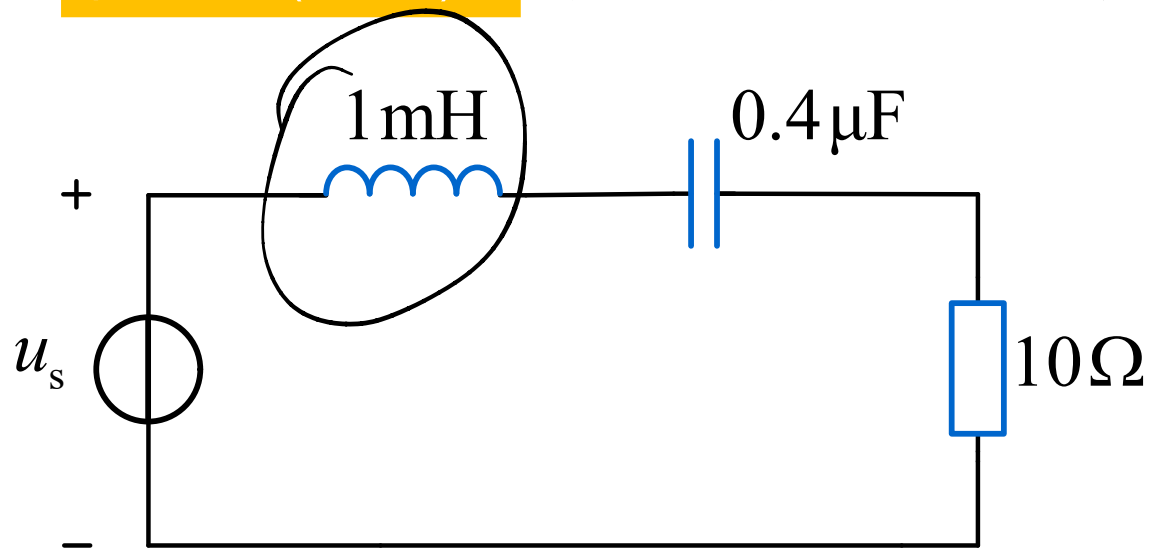
14.4 串联谐振

例题2 (基础) 求图示串联谐振电路的品质因数



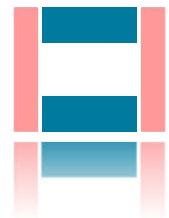
14.4 串联谐振

例题2 (基础) 求图示串联谐振电路的品质因数



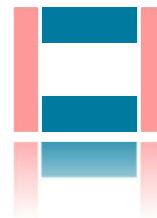
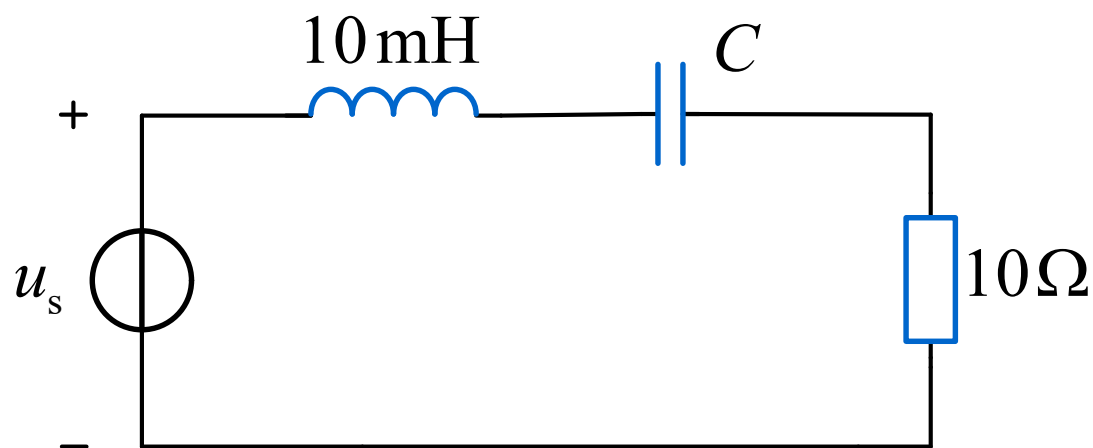
$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-3} \times 0.4 \times 10^{-6}}} = 5 \times 10^4 \text{ rad/s}$$

$$Q = \frac{\omega_0 L}{R} = \frac{5 \times 10^4 \times 10^{-3}}{10} = 5$$



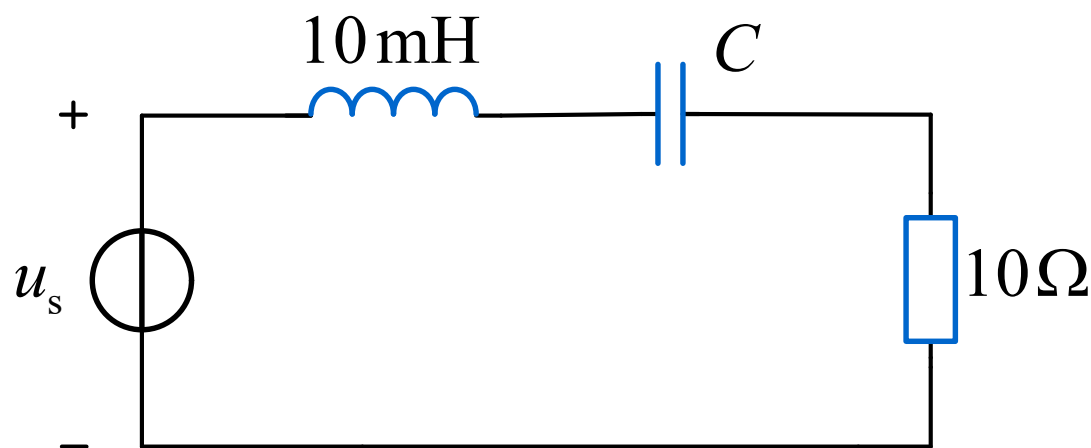
14.4 串联谐振

同步练习题2（基础） 已知图示串联谐振电路的品质因数为10，求电容 C



14.4 串联谐振

同步练习题2（基础） 已知图示串联谐振电路的品质因数为10，求电容 C



$$Q = \frac{1}{R\omega_0 C} = 10$$

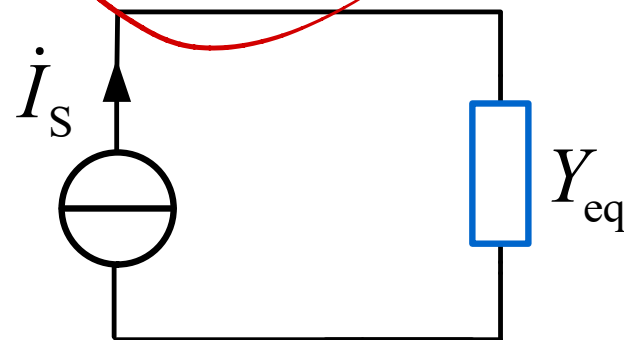
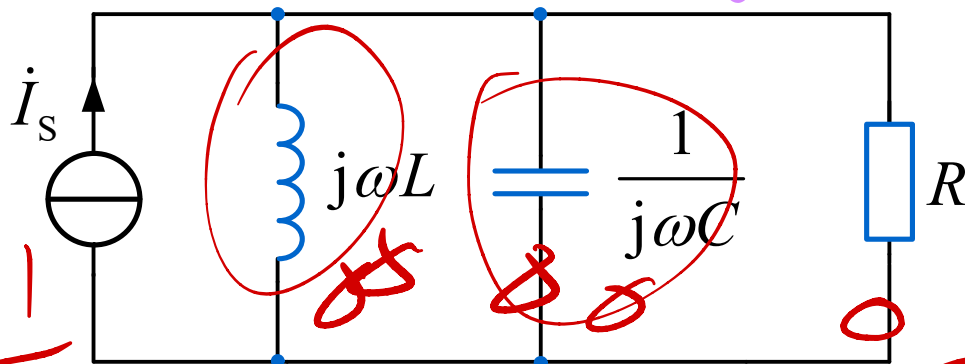
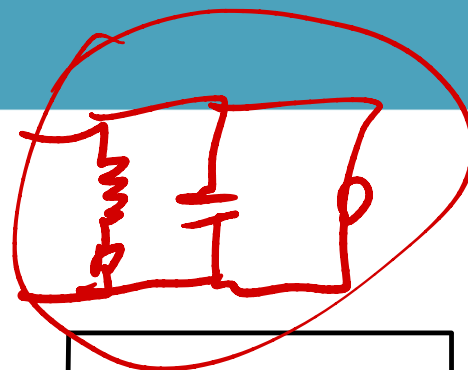
$\omega_0 = \frac{1}{\sqrt{LC}}$

答案： $C = 1\mu\text{F}$

14.5 并联谐振

并联谐振的条件

对偶



$$Y_{eq} = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

并联谐振条件: $\text{Im}(Y_{eq}) = 0$

$$Z_{eq} = \frac{1}{Y_{eq}}$$

$$= 0 \Rightarrow \omega C - \frac{1}{\omega L} = 0$$

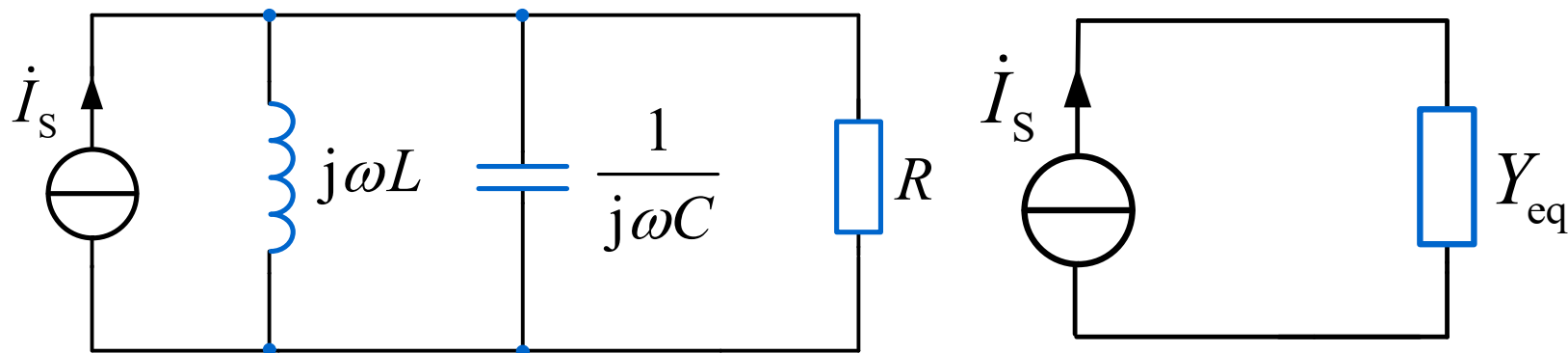
$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}, \text{ 称为并联谐振角频率}$$

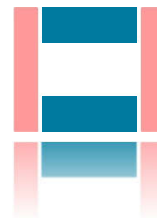
谐振 L、C 阻抗无穷大

$$\omega^2 LC = 1$$

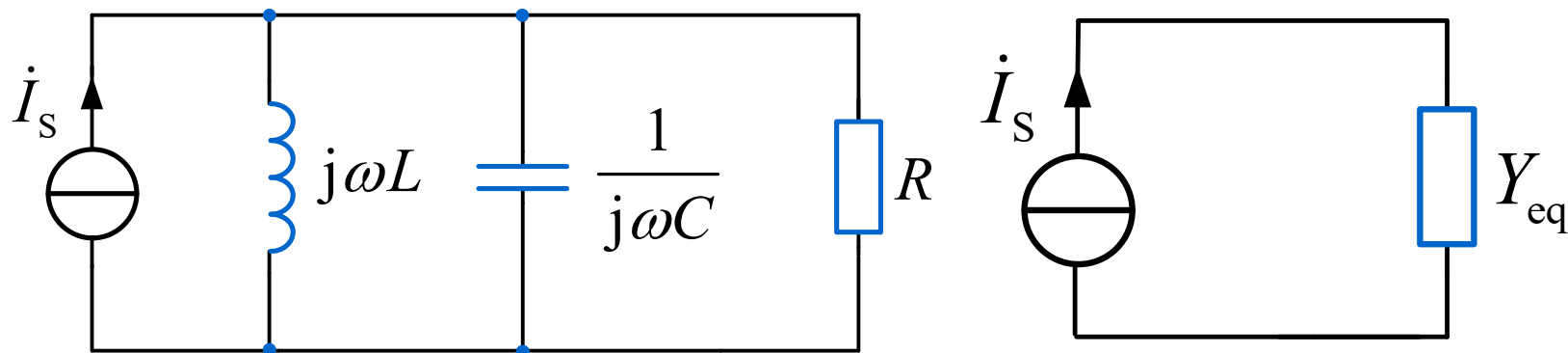
14.5 并联谐振



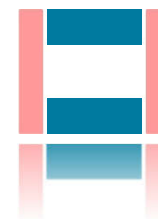
并联谐振的特点： δ 串谐振



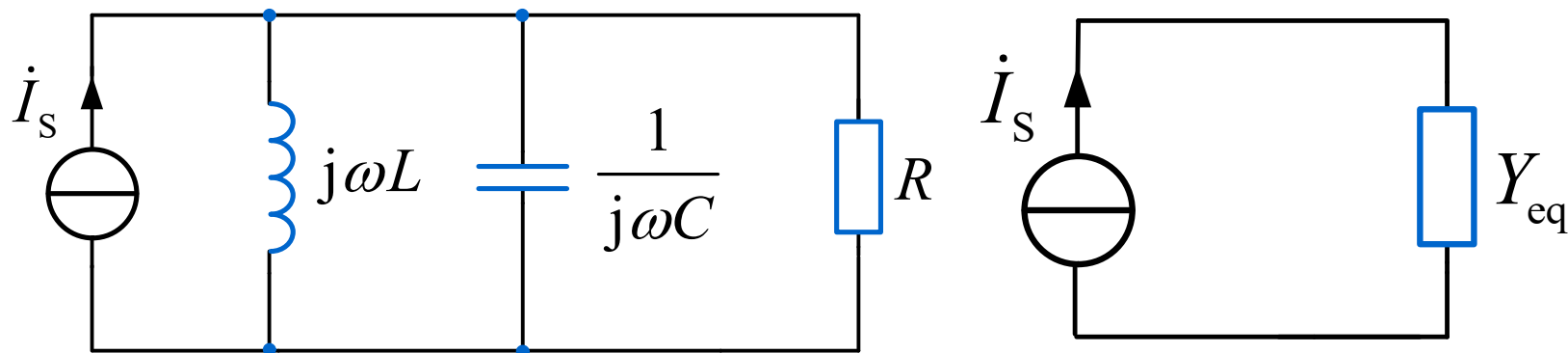
14.5 并联谐振



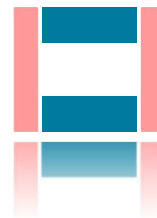
并联谐振的特点：



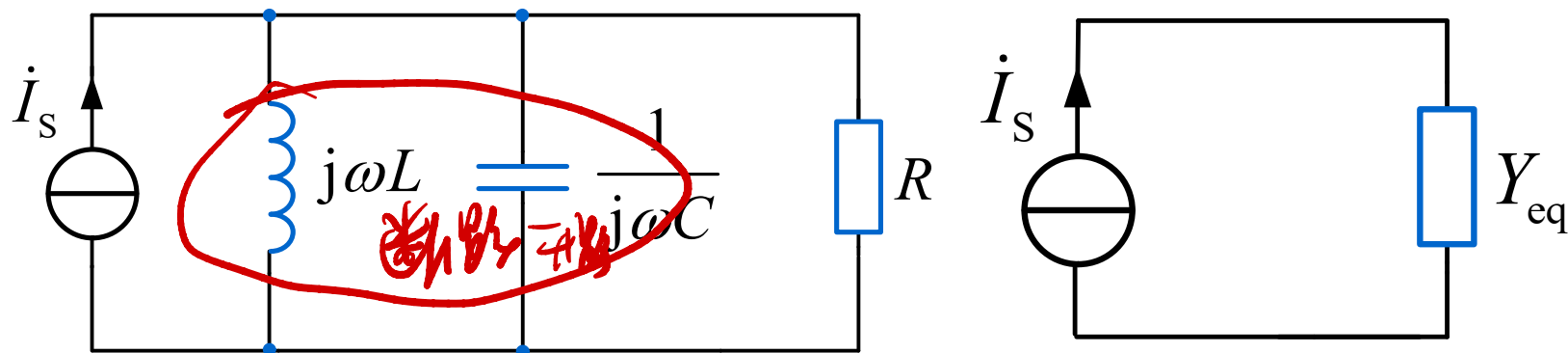
14.5 并联谐振



并联谐振的特点：



14.5 并联谐振



并联谐振的特点：

(1) 在关联参考方向下，等效阻抗（为纯电导）的电压与电流同相位，电路功率因数为1，无功功率为0

(2) 等效导纳的模值最小 $|Y|_{\min} = 1/R$

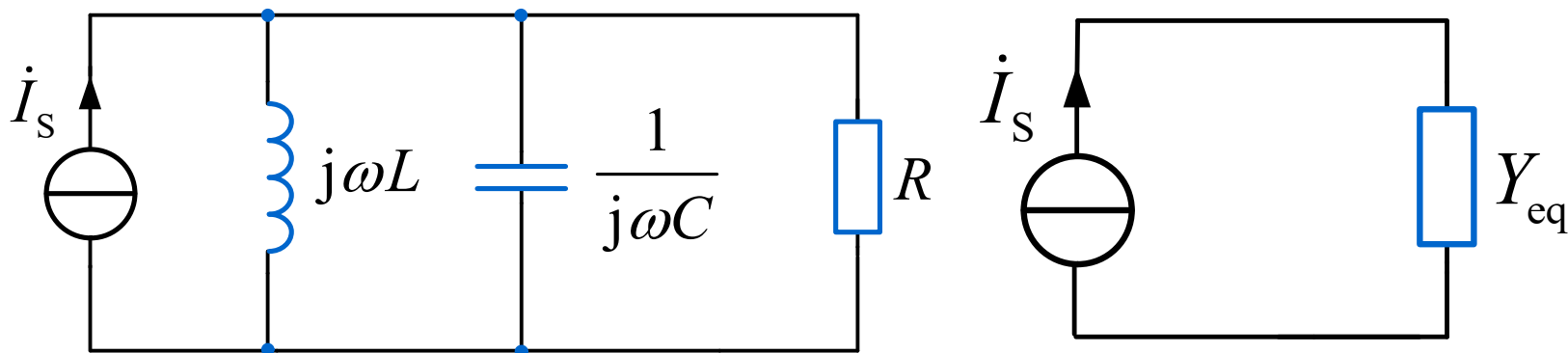
(3) 电阻电流有效值最大 $I_{R\max} = \frac{(1/R)I_s}{|Y|_{\min}} = \frac{(1/R)I_s}{(1/R)} = I_s$

最大电流

电导
市阻抗



14.5 并联谐振



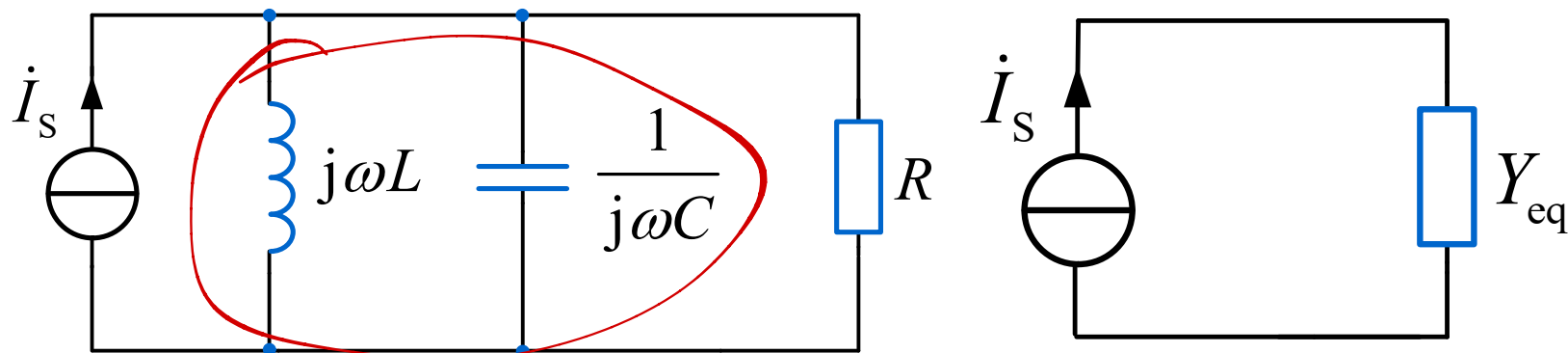
并联谐振的特点：

(4) 在关联参考方向下，等效阻抗（为纯电导）的电压与电流同相位，电路功率因数为1，无功功率为0

(2) 等效导纳的模值最小 $|Y|_{\min} = 1/R$

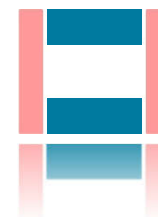
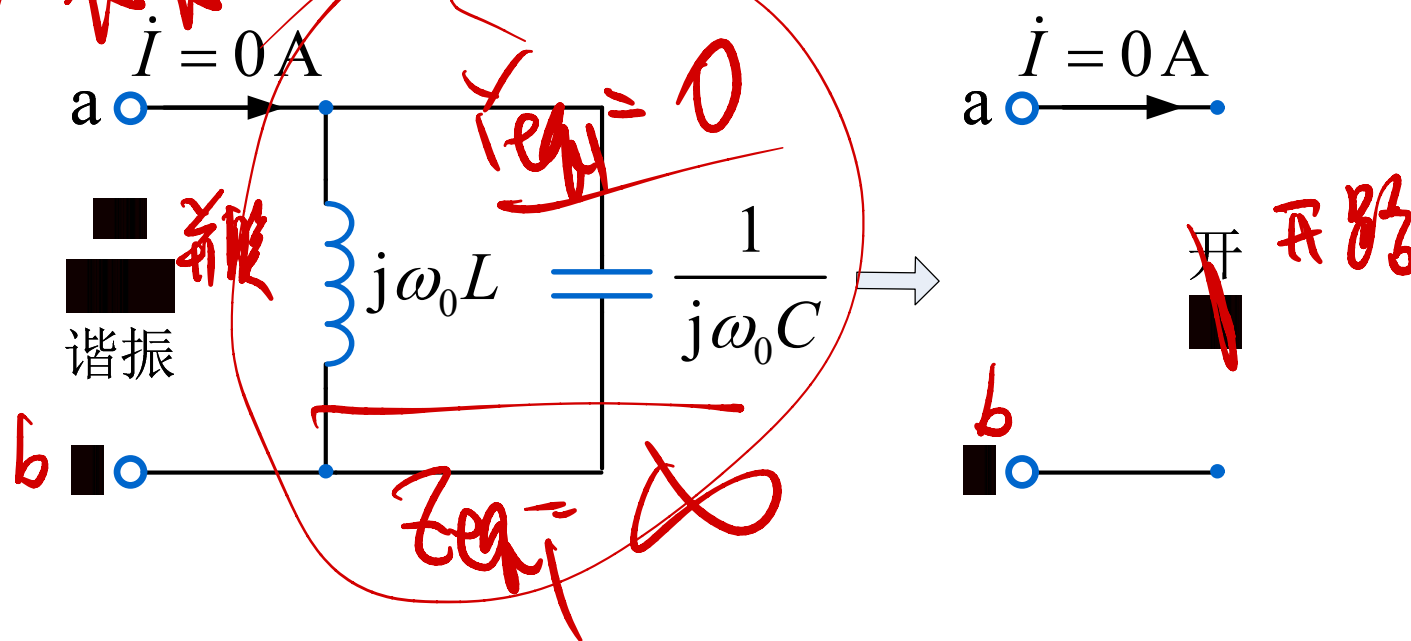
(3) 电阻电流有效值最大 $I_{R\max} = \frac{(1/R)I_s}{|Y|_{\min}} = \frac{(1/R)I_s}{(1/R)} = I_s$

14.5 并联谐振

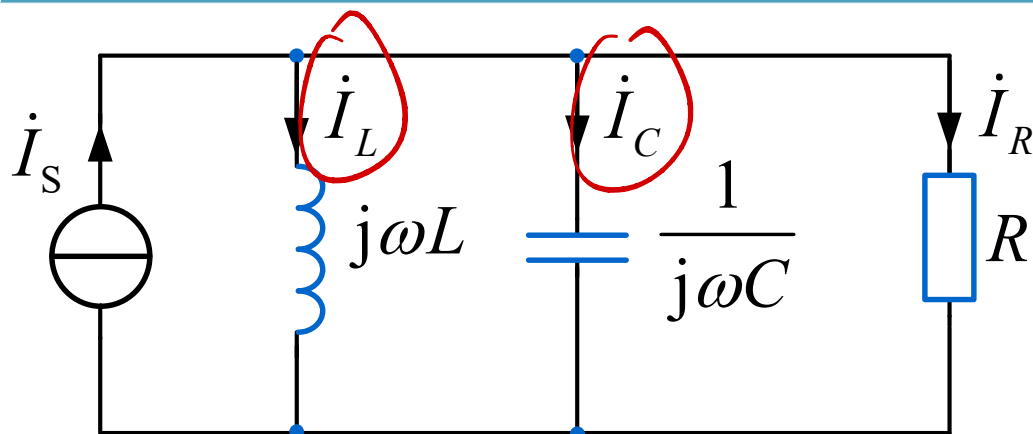


并联谐振的特点：

(1) LC并联谐振的等效导纳为0，相当于开路



14.5 并联谐振



$$\dot{I}_L \neq 0$$

$$\dot{I}_C \neq 0$$

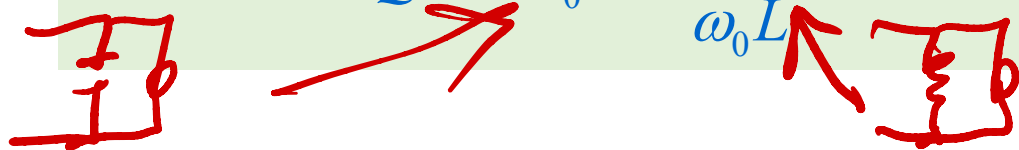
$$\dot{I}_L + \dot{I}_C = 0$$

并联谐振的特点：

(5) 电感电流和电容电流互为相反数，有效值相等，可能超过电流源电流有效值。

$$\dot{I}_L + \dot{I}_C = 0 \quad I_L = I_C = \left| \frac{1/j\omega_0 L}{Y_{eq}} \dot{I}_s \right| = \frac{R}{\omega_0 L} I_s = \left| \frac{j\omega_0 C}{Y_{eq}} \dot{I}_s \right| = \underline{R\omega_0 C I_s}$$

当品质因数 $Q = R\omega_0 C = \frac{R}{\omega_0 L} > 1$ 时，电容和电感出现过电流

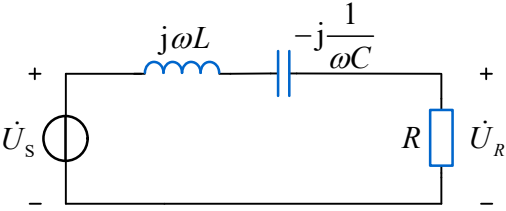
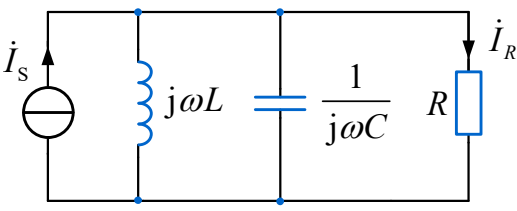


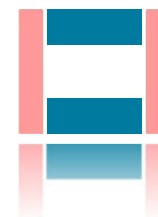
谐振

$Q \gg 1$

14.5 并联谐振

串联谐振和并联谐振特点总结

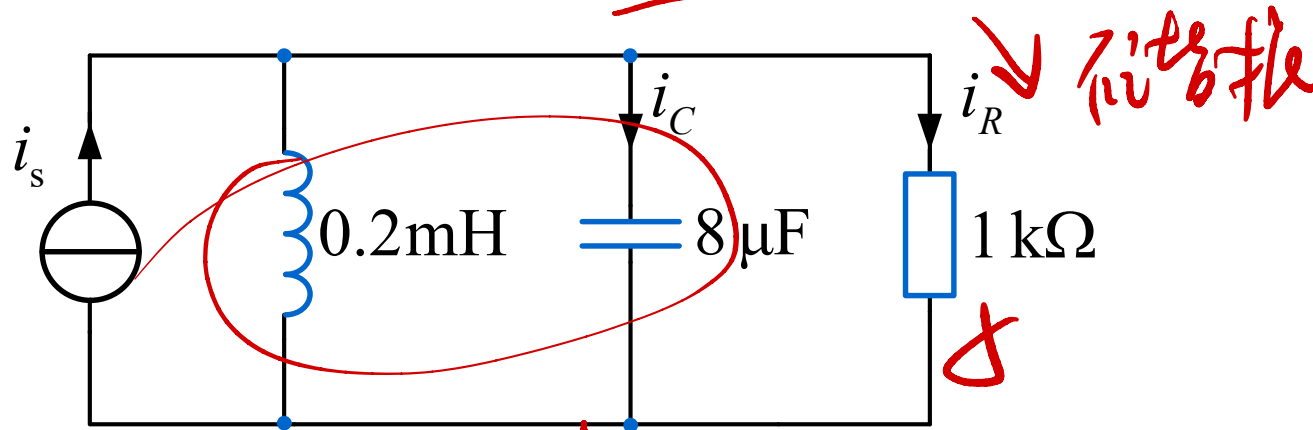
串联谐振特点	并联谐振特点
 RLC 与电压源串联	 RLC 与电流源并联
等效阻抗为纯电阻，阻抗虚部 $\text{Im}(Z) = 0$	等效导纳为纯电导，导纳虚部 $\text{Im}(Y) = 0$
谐振频率 $\omega_0 = \frac{1}{\sqrt{LC}}$	谐振频率 $\omega_0 = \frac{1}{\sqrt{LC}}$
在关联参考方向下，等效阻抗的电压与电流同相位	在关联参考方向下，等效导纳的电流与电压同相位
功率因数为最大值 1	功率因数为最大值 1
无功功率为 0 var	无功功率为 0 var
等效阻抗模值最小，电阻电压有效值最大	等效导纳模值最小，电导电流有效值最大
LC 并联等效阻抗为 $0\ \Omega$ ，相当于短路	LC 并联等效导纳为 $0\ \text{S}$ ，相当于开路
电感电压和电容电压有效值相等，可能过压	电感电流和电容电流有效值相等，可能过流
品质因数 $Q = \frac{\omega_0 L}{R} = \frac{1}{R\omega_0 C}$	品质因数 $Q = \frac{R}{\omega_0 L} = R\omega_0 C$



14.5 并联谐振

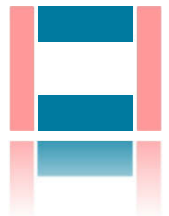
例题3 (基础)

图示正弦交流电路电流源电流有效值为1 mA。求电路的并联谐振角频率并分别计算谐振时和 $\omega = 2 \times 10^4 \text{ rad/s}$ 时的电阻电流有效值。



$$\omega_0 = \frac{1}{\sqrt{LC}} = 2.5 \times 10^4 \text{ rad/s}$$

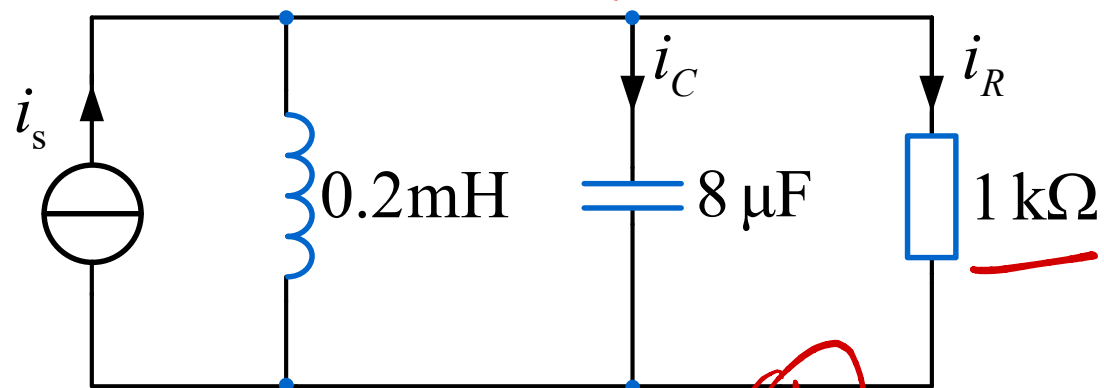
谐振时 $I_R = I_S = 1 \text{ mA}$



14.5 并联谐振

例题3 (基础)

图示正弦交流电路电流源电流有效值为1 mA。求电路的并联谐振角频率并分别计算谐振时和 $\omega = 2 \times 10^4 \text{ rad/s}$ 时的电阻电流有效值。



并联谐振
与串联谐振

$$I_p = \frac{\frac{1}{1000}}{\frac{1}{1000} + j\omega L + j\omega C}$$

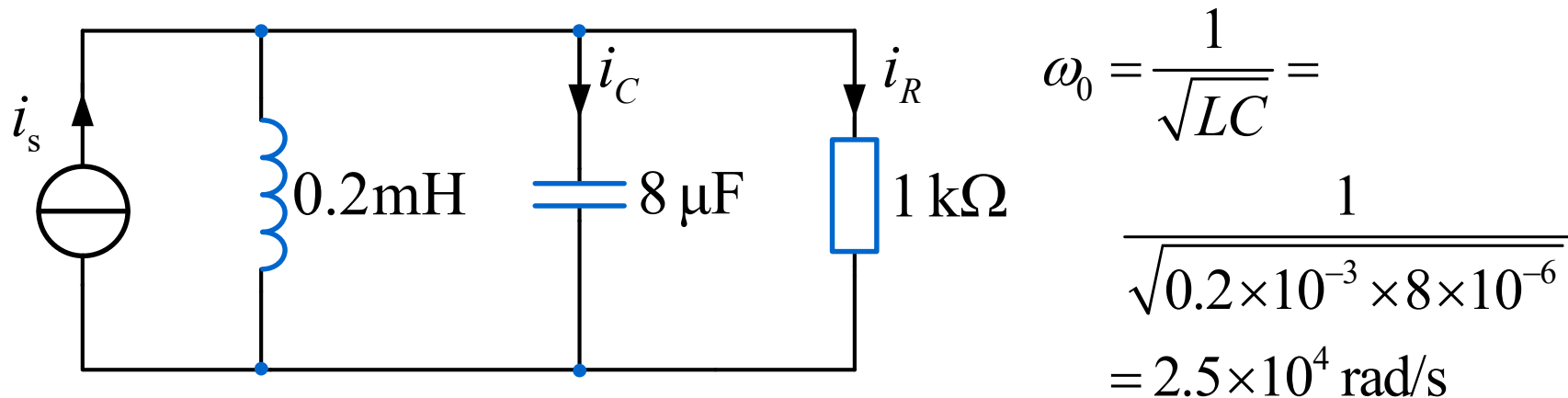
2×10^4

$$I_s = \frac{\frac{1}{1000}}{\sqrt{\left(\frac{1}{1000}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

14.5 并联谐振

例题3 (基础)

图示正弦交流电路电流源电流有效值为1 mA。求电路的并联谐振角频率并分别计算谐振时和 $\omega = 2 \times 10^4$ rad/s 时的电阻电流有效值。



$\omega = 2 \times 10^4$ rad/s 时

谐振时 $I_R = I_s = 1 \text{ mA}$

0.011 < 1

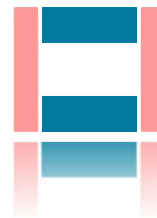
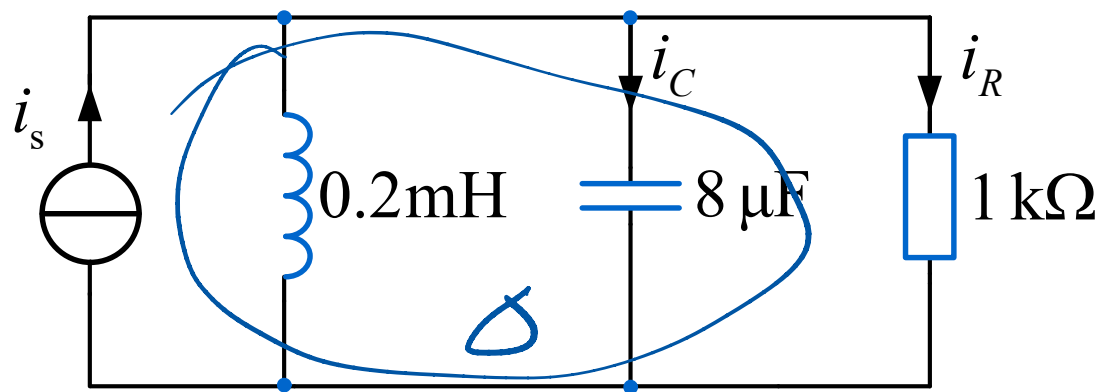
$$I_R = \left| \frac{\frac{1}{R}}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} \right| i_s = \left| \frac{10^{-3}}{10^{-3} + \frac{1}{j2 \times 10^4 \times 0.2 \times 10^{-3}} + j2 \times 10^4 \times 8 \times 10^{-6}} \right| \times 1 \approx 0.011 \text{ mA}$$

0.011 mA

14.5 并联谐振

同步练习题3（基础）

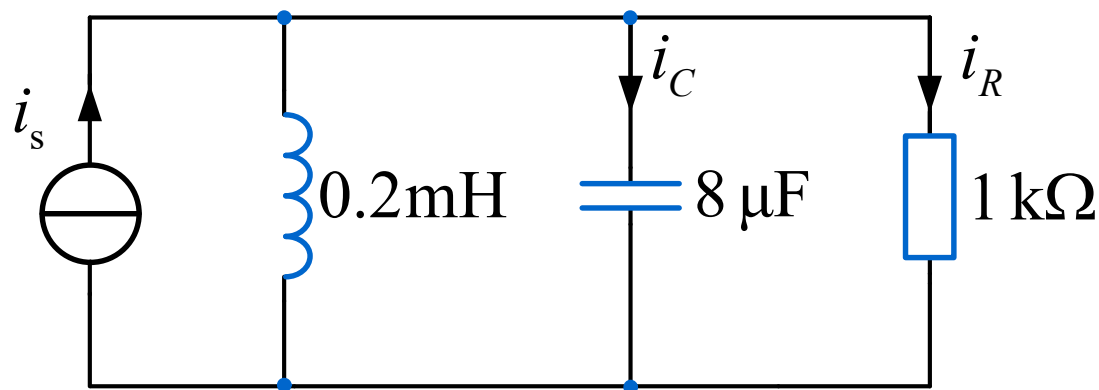
图示正弦交流电路中的电流源电流有效值为1 mA。求电路发生并联谐振时电容电流有效值。



14.5 并联谐振

同步练习题3（基础）

图示正弦交流电路中的电流源电流有效值为1 mA。求电路发生并联谐振时电容电流有效值。



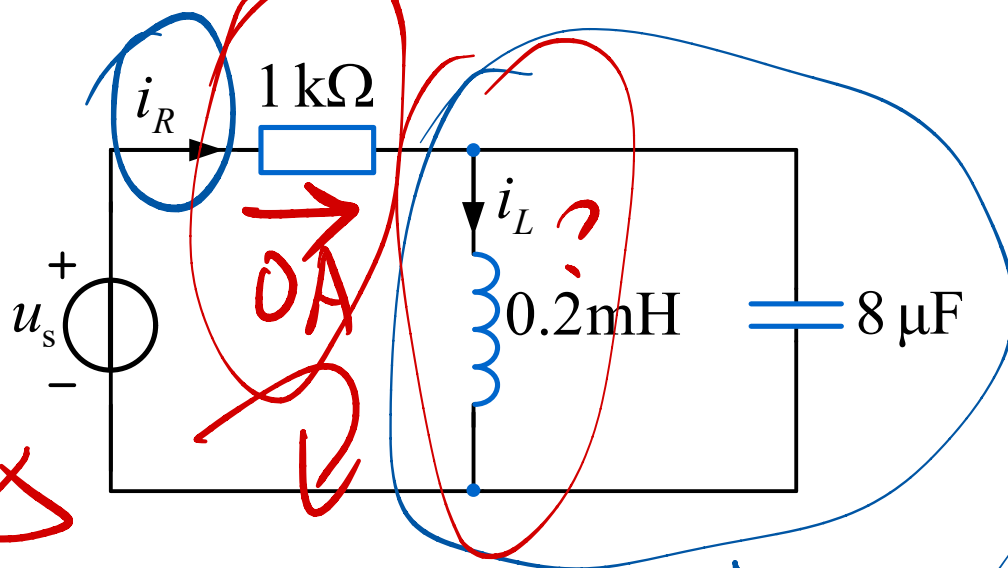
过流

答案: $I_C = 200\text{mA}$

14.5 并联谐振

例题4 (提高)

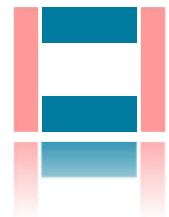
图示正弦交流电路电压源电压有效值为1 V。当电路发生谐振时，求电阻电流有效值和电感电流有效值。



$LC \rightarrow \infty \quad \omega = \frac{1}{\sqrt{LC}} \checkmark$

$Z_{eq} = \infty$
 $\left(\frac{1}{Z_L} - \left(\frac{\dot{U}_s}{j\omega L} \right) \right)$

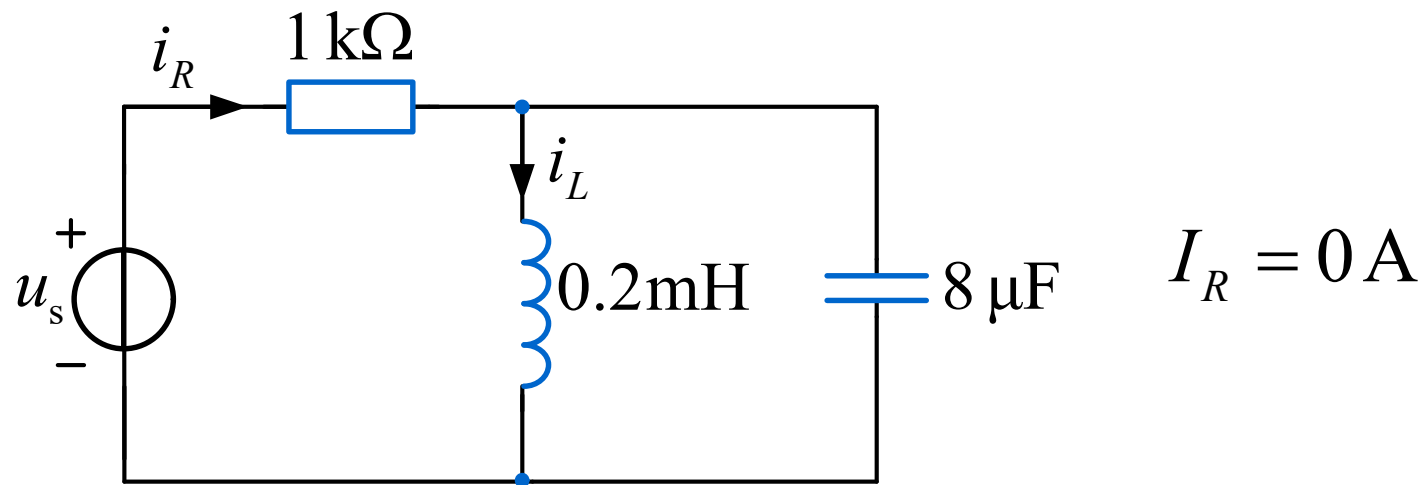
$Y_{eq} = 0$



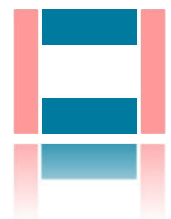
14.5 并联谐振

例题4 (提高)

图示正弦交流电路电压源电压有效值为1 V。当电路发生谐振时，求电阻电流有效值和电感电流有效值。



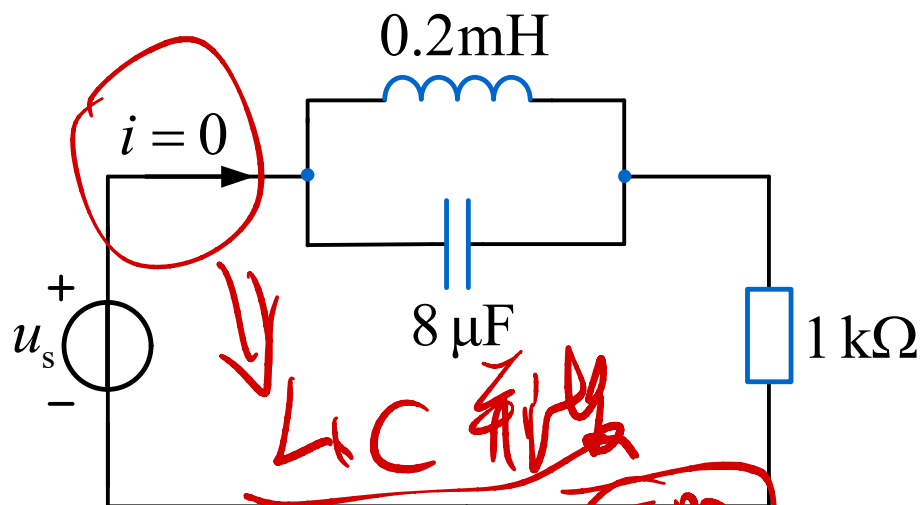
$$I_L = \left| \frac{\dot{U}_s}{j\omega_0 L} \right| = \frac{U_s}{\frac{1}{\sqrt{LC}} L} = \frac{U_s}{\sqrt{\frac{L}{C}}} = \frac{1}{\sqrt{\frac{0.2 \times 10^{-3}}{8 \times 10^{-6}}}} = 0.2\text{ A}$$



14.5 并联谐振

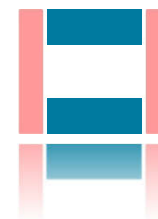
同步练习题4（提高）

图示正弦交流电路电压源电压有效值为1 V。求电容的无功功率。



$$Q = \frac{1}{\sqrt{LC}}$$

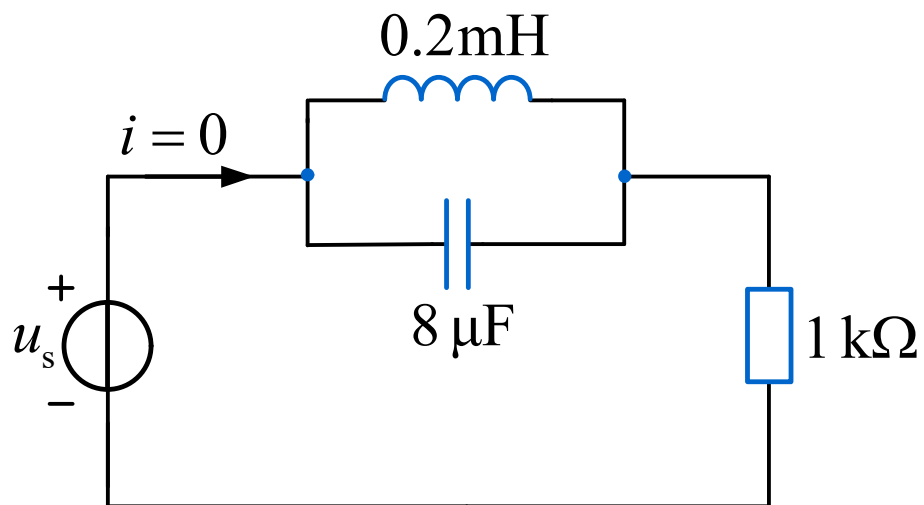
逆问题



14.5 并联谐振

同步练习题4（提高）

图示正弦交流电路电压源电压有效值为1 V。求电容的无功功率。



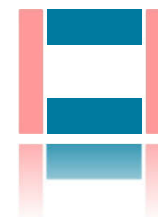
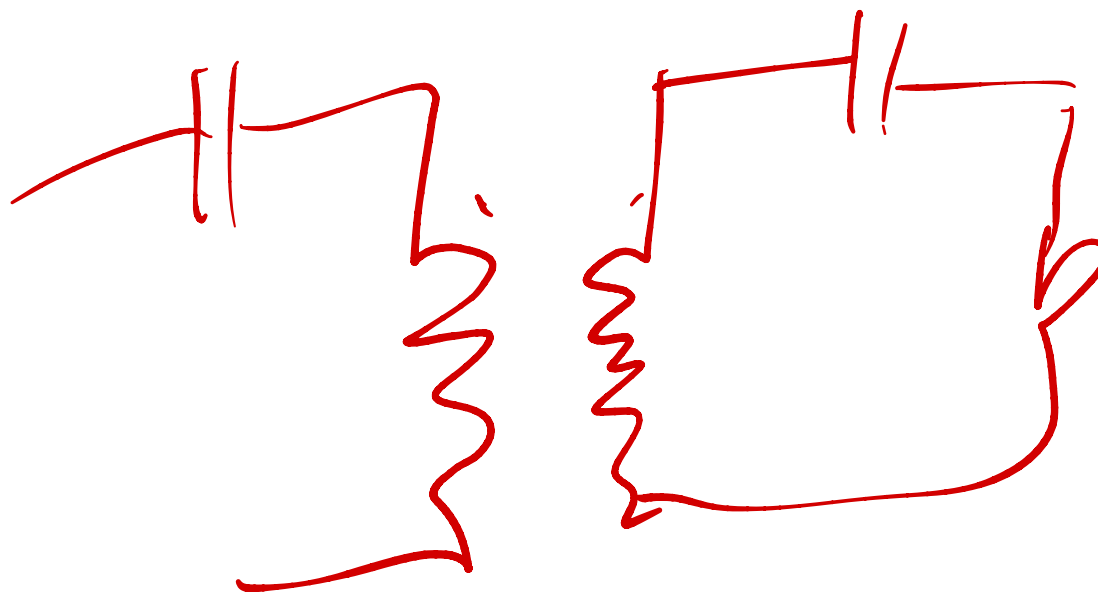
答案: $Q_C = -0.2\text{ var}$

14.3-14.5 谐振——谐振的应用

谐振具有很多非常特殊的特点，因此应用非常广泛。

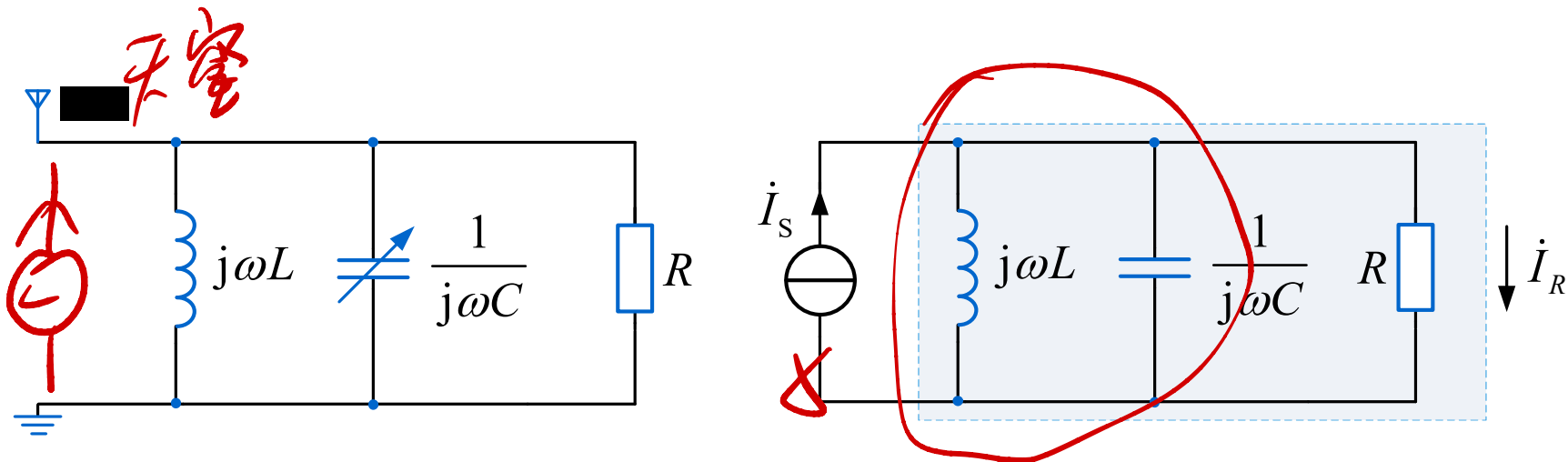
例如，收音机调频、无线电能传输、产生试验用高电压等。

此处以收音机调频为例，介绍谐振如何应用于实际。



14.3-14.5 谐振——谐振的应用

收音机调频

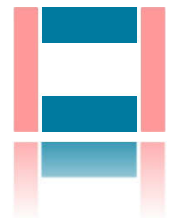
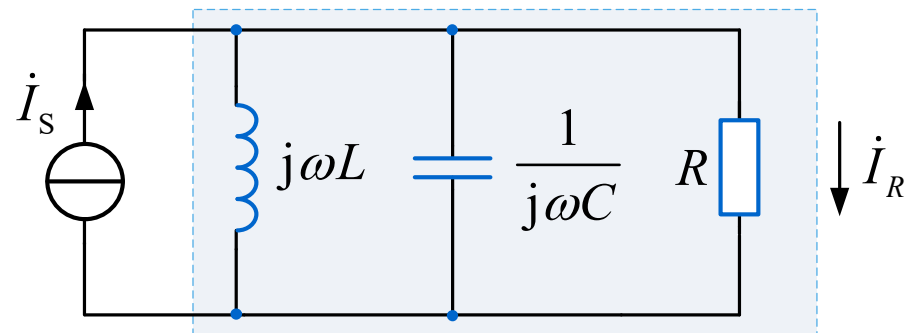
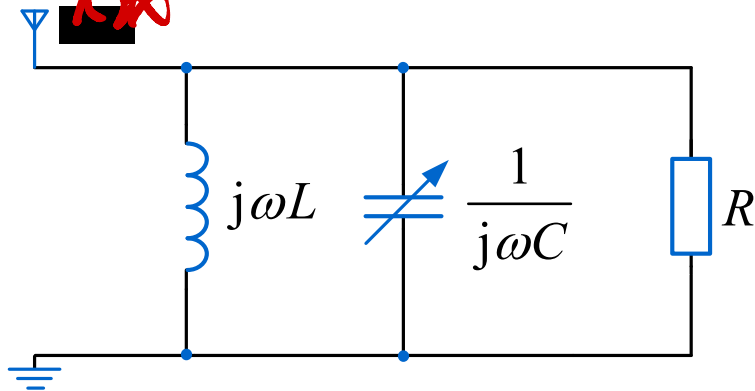


- 通过调节可变电容的电容值，可以使电路对想收听电台信号的频率发生并联谐振。此时电阻电流有效值最大，收听到的信号最强。
- 其他电台不同频率的信号同样也会通过天线进入收音机电路，不过不会发生并联谐振，因此电阻电流有效值一般比并联谐振时的电阻电流有效值小得多，信号也弱的多，不会干扰想收听电台的信号。

14.3-14.5 谐振——谐振的应用

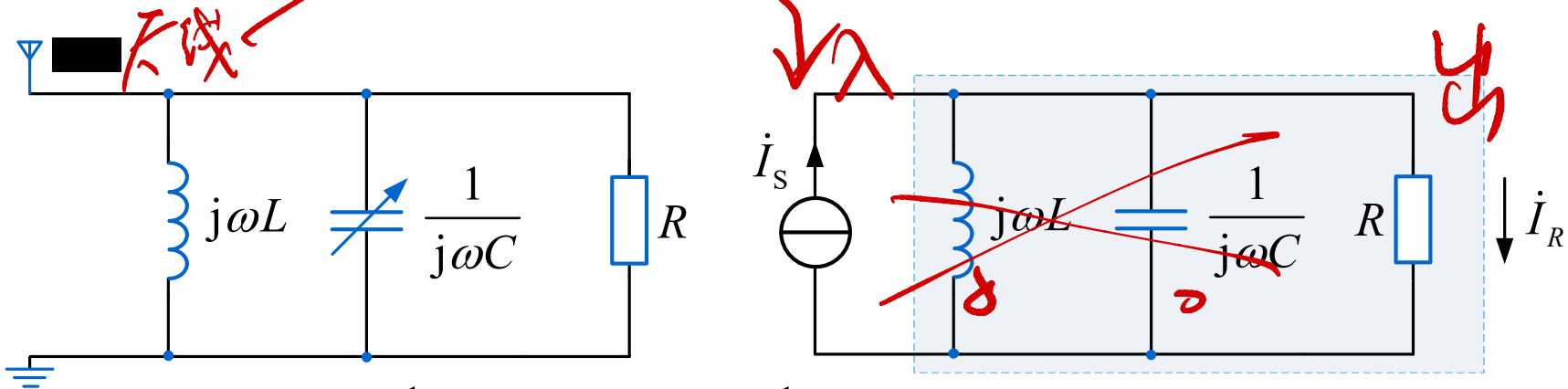
收音机调频

天线



14.3-14.5 谐振——谐振的应用

收音机调频



$$H(j\omega) = \frac{\dot{I}_R}{\dot{I}_s} = \frac{\frac{1}{R}}{\frac{1}{R} + \frac{1}{j\omega L} + j\omega C} = \frac{\frac{1}{R}}{\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)}$$

$$M(j\omega) = |H(j\omega)| = \frac{1}{\sqrt{1 + \left(R\omega C - \frac{R}{\omega L}\right)^2}} = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}}$$

Magnitude

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

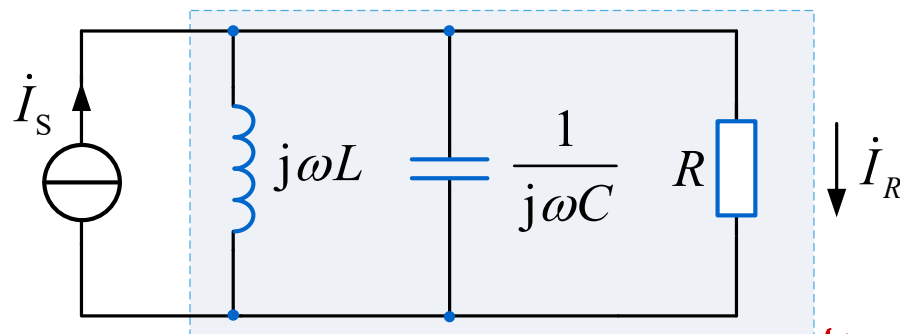
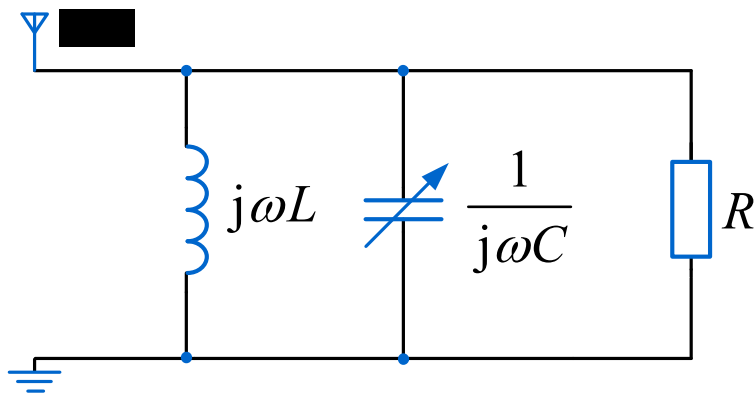
$$Q = R\omega_0 C$$

并联谐振时
 I_R 最大

品质因数

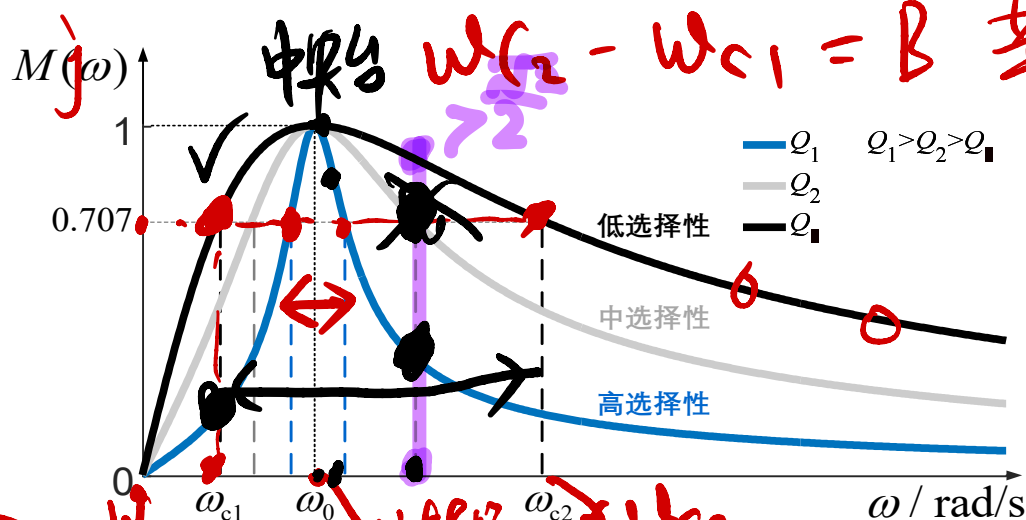
14.3-14.5 谐振——谐振的应用

收音机调频



$$M(\omega) = |H(\omega)|$$

$$= \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$



$$\text{令 } \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}} = \frac{\sqrt{2}}{2}$$

$$\text{带宽: } B = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q}$$

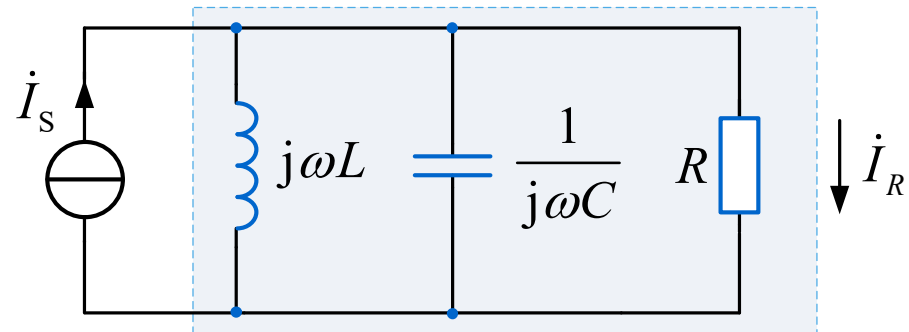
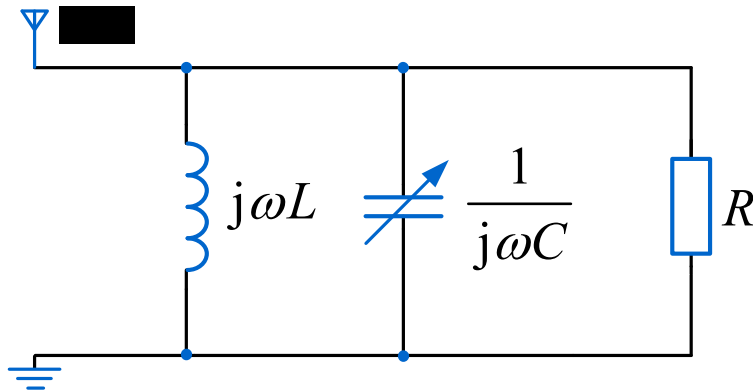
品质因数越大,

带宽越窄 频率选择性越好, 但过犹不及

$\Rightarrow$ 收音机调频 ω_{c1} ω_{c2}

14.3-14.5 谐振——谐振的应用

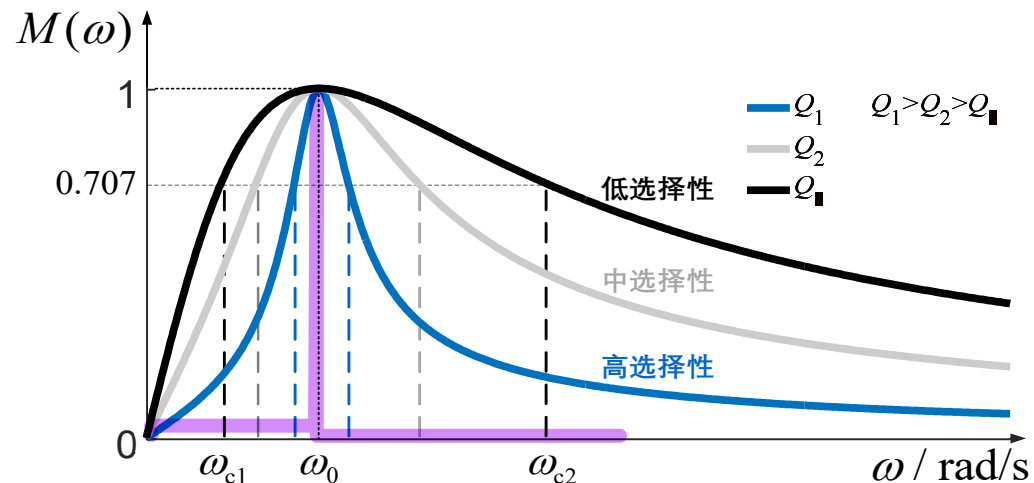
收音机调频



$$M(\omega) = |H(\omega)|$$

$$= \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

$$\text{令 } \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}} = \frac{\sqrt{2}}{2}$$



带宽: $B = \omega_{c2} - \omega_{c1} = \frac{\omega_0}{Q}$, 品质因数越大,

带宽越窄, 频率选择性越好, 但过犹不及

14.3-14.5 谐振——小结

- 串联谐振的条件是阻抗虚部为零
- 并联谐振的条件是导纳虚部为零
- 串联谐振和并联谐振的特点详见表格总结，务必理解记住！
- 谐振的特点可以广泛应用于实际
- 收音机调频的带宽与品质因数成反比 带宽： $B = \frac{\omega_0}{Q}$

感谢大家聆听

主讲人：邹建龙

时 间： 年 月 日

